1. Introduction

ACOLITE bundles the atmospheric correction algorithms and processing software developed at RBINS for aquatic applications of metre and decametre satellite data, from among others Landsat (5/7/8/9) and Sentinel-2 (A/B), Pléiades and PlanetScope. ACOLITE is in particular suited for processing of turbid waters and small inland water bodies, but can be used with reasonable success over clearer waters and land.

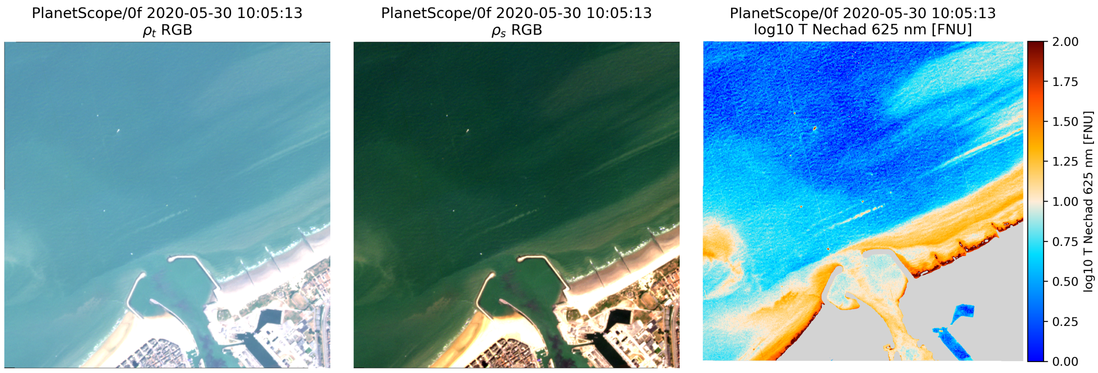

Figure 1: PlanetScope image of Oostende 2020-05-30 centred on the RT1 measurement platform. Left: ρ_t top-of-atmosphere RGB composite. Middle: ρ_s surface-level RGB composite. Right: Nechad turbidity as retrieved from the 625 nm Red band.

ACOLITE performs both the atmospheric correction and can output several parameters derived from water reflectances. RGB composites and PNG maps can also be generated. ACOLITE was originally implemented in IDL (2014–2017), and has been translated into Python (2018–2021). A new generic Python version (2021–present) was released for public beta in April 2021 that integrates the processing of metre-scale sensors such as Pléiades, WorldView, and PlanetScope, as well as the processing of Sentinel-3 (A/B) OLCI data. The original IDL and Python implementations are no longer supported or developed. Version 20210114.0 of the binary release is the last version based on the older Python codebase, versions after are based on the new generic Python codebase.

Please visit the ACOLITE forum (http://odnature.naturalsciences.be/remsem/acolite-forum/) or email Quinten (quinten.vanhellemont@naturalsciences.be) for bug reports, support and feature requests. The latest ACOLITE Python source code is available from GitHub (https://github.com/acolite/acolite). The previous Landsat and Sentinel-2 version, and the metre-scale version, are no longer supported but both versions remain available on GitHub (respectively at https://github.com/acolite/acolite_ls2 and https://github.com/acolite/acolite_mr).

2. Atmospheric correction

2.1. Overview

ACOLITE Python includes two atmospheric correction algorithms, the default "Dark Spectrum Fitting" or DSF algorithm (Vanhellemont and Ruddick, 2018; Vanhellemont, 2019a, 2020c) and the older "Exponential extrapolation" or EXP algorithm (Vanhellemont and Ruddick, 2014, 2015, 2016). The use of the DSF is recommended, but the EXP is included for completeness, and for users and applications that rely on it. The adaptation and applicability of the DSF to Landsat and Sentinel-2 is described in Vanhellemont (2019a, 2020c). DSF for metre-scale imagery can be found in Vanhellemont and Ruddick (2018); Vanhellemont (2019b, 2020c, 2023), and the adaptation to Sentinel-3 OLCI is discussed in Vanhellemont and Ruddick (2021). Braga et al. (2022) presented first results of ACOLITE/DSF processing of hyperspectral PRISMA data.

The Thermal Atmospheric Correction Tool (TACT, Vanhellemont, 2020a,b) is now integrated in ACOLITE and can be used for retrieving surface temperatures from the Landsat sensors. Additional TACT validation is provided for near-shore waters in Vanhellemont et al. (2022) and for Antarctic mountain sides in Vanhellemont et al. (2021). Some additional configuration is required for running TACT as it relies on atmospheric profile inputs and libRadtran (see further).

2.2. DSF options

2.2.1. Aerosol correction

The DSF computes aerosol optical depth (τ_a at 550 nm) based on multiple dark targets in the scene or subscene, with no a priori defined dark band. For each band the darkest object is either estimated from (1) the absolute minimum, (2) a percentile, or (3) the offset from an OLS fit to the first thousand pixels in the histogram, and a "dark spectrum", ρ_{dark} is created. A Continental or Maritime aerosol model is chosen based on the lowest RMSD between the observed ρ_{dark} and the retrieved ρ_{path} for the two closest fitting bands. Four settings for the DSF are implemented:

1. A "fixed" τ_a option, which computes a single τ_a for the scene or subscene if limits for a ROI are provided.
2. A "tiled" τ_a option, which divides the full scene in approximately 6x6 km tiles, retrieves τ_a per tile, and interpolates retrieved atmospheric parameters to the full scene.
3. A "segmented" τ_a option, which computes a single fixed τ_a per lake polygon in the scene/subscene, see polylakes keyword.
4. An experimental "resolved" τ_a setting, which treats each pixel individually. This option is currently only included for development purposes. Note that processing time is much longer for this option.

2.2.2. Glint correction

The DSF now can include a wind speed based air-water interface correction (see Vanhellemont, 2020c; Vanhellemont and Ruddick, 2021) that is disabled by default. For viewing geometries close to the principal sun glint spot this correction is rather sensitive to the wind speed, and may provide poor results when rather coarse ancillary wind data is used. High resolution imagery often has spatially resolved sun glint, with distinct patterns varying at short length scales, which cannot be corrected using such an approach. If SWIR channels are available, or NIR channels if the water is sufficiently clear, an estimate of the glint can be made and extrapolated to the shorter wavelength channels.

The DSF uses the bands and pixels giving the lowest estimate of aerosol optical thickness over the processed region or tile. This allows for an estimation of the atmospheric path reflectance that is relatively insensitive to sun glint. The sun glint signal will however still be present in the resulting surface reflectance. A sun glint correction option is included, which by default uses the SWIR bands to estimate the glint signal, as in Harmel et al. (2018). The implementation and performance is discussed in Vanhellemont (2019a). An alternative sun glint correction is provided, which estimates the average glint reflectance in a range of SWIR bands, and extrapolates this to the VNIR bands using a modeled reflectance shape. The alternative glint correction is currently only available when processing with "fixed" aerosol optical depth estimate.

2.3. EXP options

The EXP uses Rayleigh corrected reflectance in two bands to estimate the aerosol reflectance (ρ_{am}). By default the two SWIR bands are used, and ρ_{am} is estimated in the 2.2 μm band and extrapolated to the VIS and NIR bands using an exponential function and the ratio of 1.6 μm to 2.2 μm Rayleigh corrected reflectances.

The band set can be changed (see further) to include NIR/SWIR and red/NIR options, that both use a fixed ϵ due to non-zero ρ_w in at least one of the bands. The red/NIR option needs to assume a water reflectance model, for which by default the similarity spectrum (Ruddick et al., 2006) is used. For the SWIR based correction, options are included for processing with variable and fixed ϵ and ρ_{am}. A comparison between DSF and EXP methods, showing the poor performance of the latter in the blue bands, is discussed in Vanhellemont (2019a).

2.4. Outputs

The atmospheric correction procedure generates two NetCDF files containing reflectance data: (1) Level 1, top-of-the atmosphere reflectances (ρ_t, rhot_*), L1R and (2) Level 2, surface level reflectances (ρ_s, rhos_*), L2R. Based on these files the RGB composites are generated, and derived parameters are computed. Derived parameters are output to an L2W NetCDF file. Certain sensors will generate an additional L1R file for their higher spatial resolution panchromatic channel. TACT processing generates an additional L2T file. Optionally the datasets from the NetCDF files can be output to GeoTIFF files if the original input data were in a projection supported by GeoTIFF.

2.5. Ancillary data

The use of ancillary and SRTM DEM data requires the set up of an EarthData account (https://earthdata.nasa.gov). Your EarthData account needs to have approval for the OB.DAAC Data Access and LP DAAC Data Pool respectively to access these datasets. Login to your EarthData account to check Authorised Apps, and click Approve More Applications if necessary: https://urs.earthdata.nasa.gov/profile

Your EarthData username and password have to be set in the operating system environment variables "EARTHDATA_u" and "EARTHDATA_p" respectively, or they can be provided in your ACOLITE settings file or config.txt file. The differences between using ancillary data or not were found to be very small (Vanhellemont, 2020c). The username and password can also be provided through the use of a .netrc file, machine earthdata.

2.6. LUTs

The look-up tables (LUTs) required for ACOLITE/DSF are not included in the main distributions to reduce initial download volume. Many sensors are now supported, and not every user needs to have the LUTs for all sensors locally. The LUTs should be retrieved automatically from the acolite_luts repository on GitHub when needed during processing, or when the --retrieve_luts option is used (see further). If the automated retrieval does not work for some reason, LUTs can be manually obtained from https://github.com/acolite/acolite_luts The LUTs should be put in the acolite/data/LUT directory keeping the file structure on acolite_luts. For example, to process L8_OLI data, put the files from the repository:

```
https://github.com/acolite/acolite_luts/tree/main/ACOLITE-LUT-202110/L8_OLI
https://github.com/acolite/acolite_luts/tree/main/ACOLITE-LUT-202110-Reverse/L8_OLI
https://github.com/acolite/acolite_luts/tree/main/Gas
https://github.com/acolite/acolite_luts/tree/main/RSKY-202102/L8_OLI
https://github.com/acolite/acolite_luts/tree/main/WV
```

into the acolite/data/LUT folder (note that this directory may need to be created):

```
acolite/data/LUT/ACOLITE-LUT-202110/L8_OLI
acolite/data/LUT/ACOLITE-LUT-202110-Reverse/L8_OLI
acolite/data/LUT/Gas
acolite/data/LUT/RSKY-202102/L8_OLI
acolite/data/LUT/WV
```

In the binary distribution the main acolite directories are named acolite_py_linux, acolite_py_mac, acolite_py_win, and hence the location of the LUTs should be (for Windows):

```
acolite_py_win/data/LUT/ACOLITE-LUT-202110/L8_OLI
acolite_py_win/data/LUT/ACOLITE-LUT-202110-Reverse/L8_OLI
acolite_py_win/data/LUT/Gas
acolite_py_win/data/LUT/RSKY-202102/L8_OLI
acolite_py_win/data/LUT/WV
```

3. Processor overview

ACOLITE is distributed in a compiled binary format and as Python source code through GitHub. Docker images are made available on DockerHub. ACOLITE can be used in GUI and CLI modes. The Python source code can also be imported into custom scripts. The GUI offers only limited configuration (file input/output, ROI setting, output parameters and loading/saving of settings files) and is aimed at new users. Full configuration can be done using plain text 'settings files', that are interpreted both by the GUI and CLI. The use of 'minimal' settings files is recommended, i.e. only adding the settings you want to change. Missing settings will be set to defaults automatically. It is recommended that users do not change the defaults settings file at config/defaults.txt but rather create a new processing settings file for a specific run. GUI and CLI use is described below for the binary distribution and Python source code.

3.1. ACOLITE GUI

Clickable launchers are provided for Linux/Mac (symlink to dist/acolite/acolite) and Windows (shortcut to dist/acolite/acolite.exe). On Linux you may need to select a terminal emulator for opening the acolite symlink. This can be done by right clicking and selecting Open With Other Application... The ACOLITE GUI can also be launched by opening a terminal emulator in the ACOLITE directory and running dist/acolite/acolite on Mac/Linux or dist/acolite/acolite.exe on Windows.
Figure 2 shows the main ACOLITE GUI window which offers the following configuration:

- Inputfile as directory: This box determines whether the inputfile to be selected using "Select input" is a single file (e.g. PlanetScope zipped bundle) or a directory (e.g. extracted Landsat or Sentinel-2 bundle).
- Select input: Select the input directory or file for processing. This should be either the directory containing the L1 scene data (e.g. a Landsat bundle or a Sentinel-2 .SAFE file) or a single input file.
- Select output: Select the output directory.
- Region of interest: Input the southern, northern, western and eastern boundaries (in decimal degrees) of the scene to be extracted for processing.
- Polygon: As alternative to providing SNWE boundaries, provide a polygon file with your region of interest (e.g. geojson). If both Region of interest and Polygon are empty the whole scene will be processed.
- L2W parameters: List the required output parameters here, see Section 5 for a full list. If empty, only L1R and L2R files will be generated.
- PNG outputs: Tick boxes for the generation of RGB composites using top-of-atmosphere (ρ_t) or (ρ_s) reflectances, and the generation of PNG maps of the requested L2W parameters.
- Save/Restore: Save the current settings as a text file, or restore settings from a text file.
- Run processing: Start the processing. The processing runs in a separate thread that can be stopped using the Stop processing or Exit buttons.
- Stop processing: Stop the processing.
- Exit: Quit the GUI.
- Logging output: Feedback from the GUI and processor is displayed here. Detailed logs should be output to a .txt file by the processor.

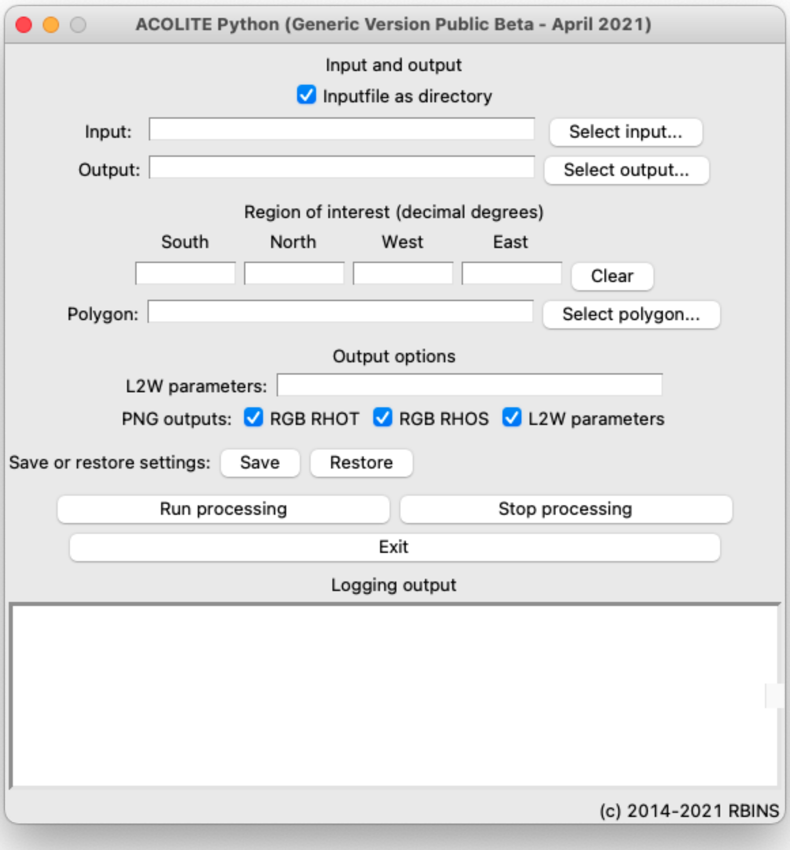

Figure 2: Screenshot from the Public Beta of the Generic ACOLITE Python (April 2021) running on Mac OS Big Sur.

3.2. ACOLITE CLI

The ACOLITE command line interface (CLI) can be launched by opening a terminal emulator and launching acolite with the –cli flag and the full path to a settings file. The following examples assume you are in the main ACOLITE directory, but ACOLITE can be launched from anywhere using its full path.

Linux and Mac:

```
dist/acolite/acolite --cli --settings=settings_file
```

Windows:

```
dist\acolite\acolite.exe --cli --settings=settings_file
```

Source code:

```
python launch_acolite.py --cli --settings=settings_file
```

Optionally a comma separated list of images (with their full paths, escape spaces with a backslash or use quotation masks around the list if needed) or a text file with paths to the images can be provided with the –inputfile flag:

Linux and Mac:

```
dist/acolite/acolite --cli --settings=settings_file --inputfile="file1,file2,file3"
```

Windows:

```
dist\acolite\acolite.exe --cli --settings=settings_file --inputfile="file1,file2,file3"
```

Source code:

```
python launch_acolite.py --cli --settings=settings_file --inputfile="file1,file2,file3"
```

The ACOLITE CLI can also be used to retrieve the LUTs required for a list of sensors before running any processing. E.g. for setting up a Docker container. In the following examples "python launch_acolite.py" can also be replaced by "dist/acolite/acolite" or "dist\acolite\acolite.exe" for the binary version.

All supported multispectral sensors:

```
python launch_acolite.py --retrieve_luts
```

Get the LUT for named sensors (here S2A/MSI and S2B/MSI):

```
python launch_acolite.py --retrieve_luts --sensor S2A_MSI,S2B_MSI
```

Get the generic LUT for processing hyperspectral sensors:

```
python launch_acolite.py --retrieve_luts --sensor hyper
```

3.3. Inputfiles

ACOLITE can process top-of-atmosphere imagery from a series of satellite sensors. Since the formatting by the commercial operators or space agencies is different for each sensor, the inputfile needs to be provided appropriately. Inputfiles can be either single files (e.g. CHRIS .hdf files or zipped Planet bundles) or directories (e.g. Sentinel .SAFE bundles or extracted Planet bundles). In this section the inputfile options are listed per sensor.

- **AMAZONIA1/CBERS4(A)** Provide the full path to a directory containing the "L2" data and metadata files per band.
- **CHRIS** Provide the full path to a CHRIS .hdf file
- **DESIS** Provide the full path to the extracted L1C bundle directory.
- **EMIT** Provide the path to the .nc file (EMIT_L1B_RAD). The OBS file (EMIT_L1B_OBS) is required to be present in the same directory as the L1B, and will be automatically detected.
- **Gaofen: GF1D / GF6** Provide the full path to the extracted L1 bundle directory containing the -MUX.tiff image and -MUX.xml metadata files.
- **HICO** Provide the full path to the L1B_ISS.nc file.
- **EO1/ALI, EO1/HYPERION** Provide the full path to the extracted L1 bundle directory.
- **Landsat 5/7/8/9** Provide the full path to the extracted bundle (i.e. extract the .tar archive). Make sure you have a L1 TOA bundle, not a L2 SR bundle. Level 1 data from both Collection1 and Collection2 are supported.
- **PlanetScope / RapidEye** Provide the path to a zipped file or unzipped bundle. Make sure you have a L1 TOA file (i.e. AnalyticMS.tif or AnalyticMS_clip.tif files) and not a SR file. For SuperDove processing order the "Analytic radiance (TOAR) - 8 band" data from the Explorer, or analytic_8b_udm2 bundles through the API.
- **Pléiades-1 A/B** Provide the path to an unzipped bundle that contains the IMG_PHR1A_MS/IMG_PHR1B_MS directories.
- **PRISMA** Provide the path to the .he5 file (PRS_L1_STD). The official L2C file (PRS_L2C_STD) is required to be present in the same directory as the L1. The L2C file does not have to be specified in your settings file, it will be automatically detected.
- **Sentinel-2** Provide the full path to the extracted .SAFE file. Make sure you have a L1C TOA bundle, not a L2 SR bundle.
- **Sentinel-3/OLCI, ENVISAT1/MERIS** Provide the full path to the extracted .SEN3 file. Make sure you have a L1 TOA bundle (S3A_OL_1_ERR or S3A_OL_1_EFR for S3A), not a L2 SR bundle (e.g. S3A_OL_2_WFR or S3A_OL_2_LFR). For MERIS processing only the 4th reprocessing .SEN3 files are supported.
- **Skysat** Provide the path to an unzipped bundle. Make sure you have a L1 TOA bundle containing analytic.tif or analytic_clip.tif files with reflectance_coefficients in the TIFFTAG_IMAGEDESCRIPTION.
- **SPOT 6/7** Provide the path to the "VOL" directory that contains the "IMG" directory (e.g. IMG_SPOT6_PMS_001_A or IMG_SPOT7_PMS_001_A). Note that this may be a few levels deep in the extracted files.
- **Venμs** Provide the path to the extracted zip file that contains the main .tif and .xml files.
- **VIIRS** Provide the path to either one of the L1B or GEO files from the I or M bands.
- **WorldView-2/3, QuickBird2** Provide the path to an unzipped bundle that contains the main .TIF and .XML files.

3.4. Example settings files

Settings files can be generated with the ACOLITE GUI, or with any text editor. Use UTF-8 encoding for best results. It is recommended to use a settings file that only includes the settings you want to change. A minimum settings file contains the full paths to the inputfile to be processed and the output directory. Processing options not provided in the settings file will be set to the defaults automatically. Comments can be added using the # or character. Comments on the same line of a setting should be stripped off, but it is recommended to put comments on the line before a setting. Several examples are given below.

A settings file that will output an area around the port of Zeebrugge in the specified directory, with water-leaving radiance reflectance in all bands output in the L2W file:

```
## Example settings for Zeebrugge
inputfile=/storage/Input/LC08_L1TP_199024_20210607_20210615_02_T1
output=/storage/Output/Zeebrugge
limit=51.3,3.0,51.4,3.25
l2w_parameters=rhow_*
```

A settings file to output Sentinel-2 data resampled to a 60 m grid for the same region of interest, increasing the masking threshold, apply a glint correction, and exporting the L2W datasets as GeoTIFF files:

```
## Example settings for Zeebrugge 60 m and GeoTIFF
inputfile=/storage/Input/S2B_MSIL1C_20200922T104649_N0209_R051_T31UES_20200924T124918.SAFE
output=/storage/Output/Zeebrugge
limit=51.3,3.0,51.4,3.25
l2w_parameters=rhow_*,tur_nechad2009
## output resolution (S2 only 10, 20, or 60 m)
s2_target_res=60
## increase default L2W masking threshold
l2w_mask_threshold=0.05
## disable wind speed based interface reflectance correction
dsf_interface_reflectance=False
## residual glint correction
dsf_residual_glint_correction=True
## output L2W to GeoTIFF
l2w_export_geotiff=True
```

A settings file for a larger region of interest, merging four Sentinel-2 tiles. Note that the scenes are listed on a single line, as a comma separated string:

```
## Example settings for Zeebrugge 60 m merging
inputfile=/storage/Input/S2B_MSIL1C_20200922T104649_N0209_R051_T31UET_20200924T124918.SAFE,
/storage/Input/S2B_MSIL1C_20200922T104649_N0209_R051_T31UES_20200924T124918.SAFE,
/storage/Input/S2B_MSIL1C_20200922T104649_N0209_R051_T31UDT_20200924T124918.SAFE,
/storage/Input/S2B_MSIL1C_20200922T104649_N0209_R051_T31UDS_20200924T124918.SAFE
output=/storage/Output/ZeebruggeMerged
limit=51.3,2.90,51.65,3.35
l2w_parameters=rhow_*,tur_nechad2009
## output resolution (S2 only 10, 20, or 60 m)
```

```
s2_target_res=60
## files given in inputfile will be merged (if from same sensor/orbit)
merge_tiles=True
```

A settings file for a larger region of interest, merging four Sentinel-2 tiles from two different UTM zones (two from zone 33, two from zone 32). Note that the zone from the first tile (in this case 33) will be used as reference to reproject the other tiles before merging:

```
## Example settings for Po 60 m merging
inputfile=/storage/Input/S2A_MSIL1C_20210331T100021_N0300_R122_T33TUL_20210331T113321.SAFE,
/storage/Input/S2A_MSIL1C_20210331T100021_N0300_R122_T33TUK_20210331T113321.SAFE,
/storage/Input/S2A_MSIL1C_20210331T100021_N0300_R122_T32TQR_20210331T113321.SAFE,
/storage/Input/S2A_MSIL1C_20210331T100021_N0300_R122_T32TQQ_20210331T113321.SAFE
output=/storage/Output/PoMerged
limit=44.6,11.9,45.3,13.1
s2_target_res=60
merge_tiles=True
```

The L1R.nc file from a previous run can also be used as an inputfile, e.g. the L1R output from the previous run:

```
## Example settings processing merged L1R file
inputfile=/storage/Output/PoMerged/S2A_MSI_2021_03_31_10_08_25_merged_L1R.nc
output=/storage/Output/NetCDF
l2w_parameters=chl_re_gons,tur_nechad2009
map_l2w=True
```

3.5. Installation from source

The Python source code is available on GitHub (https://github.com/acolite/acolite/) and includes the latest updates. The GitHub repository can be downloaded through a browser or cloned using git:

```
git clone --depth 1 https://github.com/acolite/acolite
```

To use the source code a Python 3 environment is needed with the following packages (and their dependencies):

```
numpy matplotlib scipy gdal pyproj scikit-image pyhdf pyresample netcdf4 h5py requests pygrib cartopy
```

A suitable Python environment can be set up using conda (either from miniconda or anaconda) and the packages on conda-forge:

```
conda create -n acolite -c conda-forge python=3
conda activate acolite
conda install -c conda-forge numpy matplotlib scipy gdal pyproj scikit-image pyhdf pyresample netcdf4 h5py
    requests pygrib cartopy
```

Activate the "acolite" environment before running ACOLITE:

```
conda activate acolite
```

ACOLITE can then be launched using Python, with optional command line options as described above:

```
python acolite/launch_acolite.py
```

ACOLITE can be imported in other Python scripts or Jupyter notebooks, e.g. when the git clone is in a git folder in your user home directory:

```
import sys, os, time
user_home = os.path.expanduser("~")
sys.path.append(user_home+"/git/acolite")
import acolite as ac
```

The processing can then be integrated in other scripts, e.g. by using a Python dict to provide settings:

```
settings = {"inputfile":"S2B_MSIL1C_20200922T104649_N0209_R051_T31UES_20200924T124918.SAFE",
            "output":"~/Output/",
            "s2_target_res":60,
            "dsf_residual_glint_correction":True}

ac.acolite.acolite_run(settings=settings)
```

3.6. Moving the "data" directory

The data directory can be moved to a different location if the file config/config.txt is edited appropriately to include the full path to the data directories. By default these directories are inside the acolite folder, and relative paths are used:

```
## Data directory
data_dir=data

## Scratch directory
scratch_dir=scratch

## atmospheric correction LUT data directory
lut_dir=data/LUT

## DEM SRTM HGT files
hgt_dir=data/SRTMGL3.003

## Copernicus DEM files
copernicus_dem_dir=$ACDIR/data/

## MET files
met_dir=data/MET

## SNAP directory
snap_directory=/Applications/snap

## TACT support
grid_dir=data/TACT/grid

## libRadtran directory
libradtran_dir=external/libRadtran-2.0.2
```

3.7. Setting up TACT processing

Running TACT requires libRadtran to be installed and compiled to simulate atmospheric upwelling and downwelling radiances as well as transmittances (Vanhellemont, 2020a). Inputs to libRadtran are atmospheric profiles from the ERA5 model that are retrieved from the Research Data Archive (RDA) at the University Corporation for Atmospheric Research (UCAR).

3.7.1. libRadtran

Please download libRadtran and follow compilation instructions at http://libradtran.org/doku.php TACT was successfully tested with libRadtran version 2.0.2 on Mac and Linux. The ACOLITE configuration file at acolite/config/-config.txt needs to be edited to include the full path to the libRadtran directory on your system in the libradtran_dir setting, by default this is set to external/libRadtran-2.0.2 inside the main ACOLITE directory.

To install libRadtran in the external directory (when in the main acolite directory):

```
mkdir external
cd external
wget http://www.libradtran.org/download/history/libRadtran-2.0.2.tar.gz
gzip -d libRadtran-2.0.2.tar.gz
tar -xvf libRadtran-2.0.2.tar
cd libRadtran-2.0.2
./configure
make
make check
```

3.7.2. RDA access

QV 2021-11-22 Not sure whether this "RDA access" configuration is still necessary.

And account with RDA/UCAR is required for accessing atmospheric profiles, click "Register now" at the very top of the page at (https://rda.ucar.edu/). These credentials need to be known when accessing the RDA machine, and they have to be added to your .netrc file (e.g. nano $HOME/.netrc):

```
machine rda.ucar.edu
login ADD_LOGIN_HERE
password ADD_PASSWORD_HERE
```

Your .dodsrc (e.g. nano $HOME/.dodsrc) file needs to include a full path to your .netrc file:

```
HTTP.NETRC=/path/to/.netrc
```

Note that the use of the $HOME environment variable does not work in the .dodsrc file, and that a full path needs to be given. This can be found e.g. using "find $HOME/.netrc"

3.8. Running ACOLITE in Docker

ACOLITE can be installed in a Docker container for deploying and running the processing in parallel. A working Docker environment is assumed, check out the excellent documentation at https://docs.docker.com/get-started/. The provided images and examples are based on the Ubuntu 20.04 LTS image and uses conda and conda-forge for Python environment and package management.

3.8.1. Docker images

Docker images are provided through DockerHub and will be obtained automatically when docker run is executed using either the acolite/acolite:all_latest or acolite/acolite:tact_latest tags. See Section 3.8.2 for generating your own local Docker images, or proceed to Section 3.8.3 for examples of running Docker. The DockerHub page can be found at https://hub.docker.com/r/acolite/acolite

3.8.2. Local Docker installation

It is recommended to use one of the images from DockerHub, but it is possible to make a local image. Obtain one of the examples from the GitHub (https://github.com/acolite/acolite_tools/tree/master/acolite/Docker) either with (acolite-docker-tact) or without (acolite-docker-all) libRadtran for TACT processing. Enter the docker directory (in this example acolite-docker-all), and create input and output directories:

```
cd acolite-docker-all
mkdir input
mkdir output
```

Edit the Dockerfile (e.g. "nano Dockerfile") to add/remove the sensors you want supported or not. Use a "retrieve_luts" line per sensor, or a single command with a comma separated list from the following supported sensors: S2A_MSI, S2B_MSI, L8_OLI, L9_OLI, L5_TM, L7_ETM, EO1_ALI, S3A_OLCI, S3B_OLCI, EN1_MERIS, RapidEye, SPOT6, SPOT7, PHR1A, PHR1B, VENµS_VSSC, WorldView2, WorldView3, QuickBird2, PlanetScope_0c, PlanetScope_0d05, PlanetScope_0d06, PlanetScope_0e, PlanetScope_0f, PlanetScope_22, PlanetScope_SD5, PlanetScope_SD8, Skysat1, Skysat2, Skysat3, Skysat4, Skysat5, Skysat6, Skysat7, Skysat8, Skysat9, Skysat10, Skysat11, Skysat12, Skysat13, Skysat14-Skysat19, AMAZONIA1_WFI, CBERS4A_WFI, CBERS4_WFI, GF1D_PMS, GF6_PMS, GF6_WFV

For hyperspectral sensors the "hyper" tag can be used to retrieve the generic LUTs.

After editing the Dockerfile, build the ACOLITE Docker (here using the acolite:local tag):

```
docker build --no-cache -t acolite:local .
```

3.8.3. Running Docker

Processing a scene with no custom settings can be done by mounting the image path and output paths directly and running the Docker without providing settings. The examples below uses variables for input/output/settings paths for clarity. The acolite/acolite:all_latest tag is used, which will obtain the latest version from DockerHub. If you built acolite:local yourself, use that tag. Processing a Sentinel-2 scene with default settings:

```
input=/home/quinten/Sentinel-2/Data/31UES/
    S2A_MSIL1C_20210329T105631_N0209_R094_T31UES_20210329T130721.SAFE
output=/home/quinten/Output/Default
docker run --rm -d -v $input:/input -v $output:/output acolite/acolite:all_latest
```

For running custom settings, create a settings file (e.g. "nano user_settings.txt") with your processing settings. Make sure to begin the inputfile=input/ and output=output/ settings correctly, to refer to the input/ and output/ mount points in the Docker. If you mount the scene directly as in the example above, just use inputfile=input/, if you mount a directory containing the scene, add the appropriate directory levels after inputfile=input/. For example:

```
## ACOLITE user settings for Docker
output=output/Zeebrugge
inputfile=input/Data/31UES/S2A_MSIL1C_20210329T105631_N0209_R094_T31UES_20210329T130721.SAFE
limit=51.30,3.10,51.42,3.29
```

In this case the Docker will be run with the path to the directory containing your data (in this case "Data" is a directory in "/home/quinten/Sentinel-2/") as input and a local output directory as output. The extra levels after output/ will be created. For example:

```
input=/home/quinten/Sentinel-2/Data/31UES/
    S2A_MSIL1C_20210329T105631_N0209_R094_T31UES_20210329T130721.SAFE
output=/home/quinten/Output/Default
settings=/home/quinten/user_settings.txt
docker run --rm -d -v $input:/input -v $output:/output acolite/acolite:all_latest
```

4. Processing options

This section lists most of the processing options that can be specified in a user generated settings file. Overall default settings are given in config/defaults.txt and in files per sensor in config/defaults. Sensor specific defaults may differ from the ones listed in this section.

4.1. General options

inputfile=
: Scene or comma separated list of scenes to be processed. Full paths to the scene directory (Landsat bundle or Sentinel-2 .SAFE file) have to be specified. The comma separated list has to be on a single line. Compressed files (e.g. from EarthExplorer) have to be extracted.

output=
: Directory where outputs will be generated. Will be created if it does not exist.

limit=
: Comma separated list of the bounding box coordinates of the region of interest in decimal degrees in S,W,N,E order. If empty, and polygon=None the full scene will be processed.

limit_buffer=None
: Add a buffer (decimal degrees) when determining crop for the given limit or polygon. Useful when reprojection of swath data is needed. Currently only supported for Sentinel-3 and VIIRS processing.

polygon=None
: A polygon file (e.g. geojson) specifying the bounding box for the region of interest. If None, and limit is empty, the full scene will be processed.

polygon_limit=True
: Whether to use the bounding envelope of the polygon file to crop the scene. If False, the full scene will be processed retaining data for the polygons covered by the scene. Options=True/False

polylakes=False
: Whether to use one of the global lake polygon datasets to mask out lakes in the image. Options=True/False

polylakes_database=worldlakes
: Which polylakes database to use. Options=worldlakes/hydrolakes

region_name=
: A name for the region of interest that will be appended to the output files (if not empty).

merge_tiles=False
: Controls whether inputfiles have to be merged before processing. Merging is only supported for scenes from the same orbit, and when a region of interest is specified. Options: True/False

merge_zones=True
: Controls whether inputfiles from different UTM zones have to be merged before processing. If False, a file for each UTM zone will be generated. Options: True/False

extend_region=False
: Controls whether the output file has to cover the entire limit or polygon, or whether it can be limited to the scene extent. Options: True/False

atmospheric_correction=True
: Whether to apply an atmospheric correction of not. Options: True/False

aerosol_correction=dark_spectrum
: The aerosol correction (DSF or EXP) to use. See below for algorithm specific options. Options: dark_spectrum, exponential

l2w_parameters=
: Comma separated list of the output parameters to be computed. See section 5 for a full list. If empty, only L1R and L2R files will be generated.

resolved_geometry=False
: Use per pixel resolved view/illumination geometry grids if available.

runid=
: An identifier for the current processing run. An identifier based on date/time is generated if empty.

4.1.1. Gains

gains=False

Controls the application of sensor specific TOA gains. Options: True/False

gains_toa=None

Comma separated list of gains to be used for the currently processed sensor. If not None, there has to be a comma separated list of values, one per band. Defaults can also be set per sensor in the sensor specific default files.

4.1.2. Masking

l2w_mask=True

Controls whether the L2W output parameters will be masked. Options: True/False

l2w_mask_wave=1600

Wavelength (nm) for the non-water masking. The closest band will be selected.

l2w_mask_threshold=0.0215

Threshold for the non-water masking. Pixels with ρ_t in the masking band above this threshold will be masked.

l2w_mask_water_parameters=True

Controls whether water specific parameters are to be masked. Options: True/False

l2w_mask_negative_rhow=True

Controls whether pixels with negative reflectances are to be masked. Options: True/False

l2w_mask_negative_wave_range=400,900

Wavelength range to check for negative reflectances.

l2w_mask_cirrus=True

Enable masking of cirrus clouds using a band around 1375 nm. OLI and MSI only. Options: True/False

l2w_mask_cirrus_threshold=0.005

Threshold for the cirrus masking. Pixels with ρ_t in the cirrus band above this threshold will be masked.

l2w_mask_cirrus_wave=1373

Wavelength for the cirrus masking. The closest band within 5 nm of this wavelength will be used.

l2w_mask_high_toa=True

Mask pixels with high (> l2w_mask_high_toa_threshold) top-of-atmosphere reflectance in any band. Options: True/False

l2w_mask_high_toa_threshold=0.3

Threshold for TOA reflectance masking.

l2w_mask_smooth=True

Apply Gaussian smoothing to the computed L2W mask. This option can reduce speckled masks especially in Landsat 5 and 7. Options: True/False

l2w_mask_smooth_sigma=3

Kernel size for the Gaussian smoothing of the L2W mask.

l2w_data_in_memory=True

Whether to load L2R data fully in memory when computing L2W data. When L2R data is kept into memory, L2W processing is lightly faster at the cost of memory use. L2W processing for large scenes may run out of RAM with this option enabled. Options: True/False

flag_exponent_swir=0

Bit index that will be used to store the SWIR threshold masking step (2**0).

flag_exponent_cirrus=1

Bit index that will be used to store the cirrus masking step (2**1).

flag_exponent_toa=2

Bit index that will be used to store the top-of-atmosphere threshold masking step (2**2).

flag_exponent_negative=3

Bit index that will be used to store the negative retrievals masking step (2**3).

flag_exponent_outofscene=4

Bit index that will be used to store the out-of-scene masking step (2**4).

4.1.3. Ancillary data

ancillary_data=True
Controls whether ancillary data is to be used for retrieving ozone, water vapour and atmospheric pressure. Note that for retrieving ancillary data an EarthData account is required. Options: True/False

ancillary_type=GMAO_MERRA2_MET
Ancillary file types to retrieve from the OBPG oceandata server. Option GMAO_MERRA2_MET should work in most cases and will retrieve hourly files containing pressure, ozone and water vapour. Option GMAO_FP_MET can be used for NRT processing. Other combinations: GMAO_FP_MET, TOAST,MET_NCEP,MET_NCEP_NEXT or O3_AURAOMI_24h,MET_NCEP_6h,MET_NCEP_6h_NEXT

EARTHDATA_u=
Username for your EarthData account. Can also be set as environment variable.

EARTHDATA_p=
Password for your EarthData account. Can also be set as environment variable.

uoz_default=0.3
Fixed/fallback value for ozone concentration in cm^{-1}. This value will be used if the ancillary data is disabled or unavailable and gas_transmittance=True.

uwv_default=1.5
Fixed/fallback value for water vapour concentration in g/cm^2. This value will be used if the ancillary data is disabled or unavailable and gas_transmittance=True.

pressure=None
Fixed atmospheric pressure value. Defaults to pressure_default value if equal to None and if no ancillary data is used and elevation is None.

pressure_default=1013.25
Default fixed atmospheric pressure value (1013.25 mbar). Used if pressure is None and if no ancillary data is used and elevation is None.

elevation=None
Elevation to compute atmospheric pressure according to a standard atmosphere. If pressure is provided then this value will be ignored. Set pressure=None to use provided elevation.

dem_pressure=False
Controls whether pressure is to be computed from DEM elevation. Options: True/False

dem_pressure_resolved=True
Controls whether pressure from DEM elevation is reprojected and computed per pixel or whether a scene percentile is used. Options: True/False

dem_pressure_percentile=25
Determine pressure from this percentile of DEM in the ROI/scene.

dem_pressure_write=True
Controls whether the DEM derived pressure dataset is output to the L2R NetCDF file. Options: True/-False

dem_source=copernicus30
Which DEM source to use. Currently Copernicus GLO-30, Copernicus GLO-90, SRTM and SRTM15+ are supported. SRTM has higher resolution but is limited to latitudes less than 64 degrees. Options: copernicus30, copernicus90, COP-DEM_GLO-30-DGED__2021_1, COP-DEM_GLO-30-DGED__2022_1, COP-DEM_GLO-90-DGED__2021_1, COP-DEM_GLO-90-DGED__2022_1, strm, srtm15plus

scene_download=False
Controls whether to attempt to download a Sentinel-2 or Sentinel-3 scene from the CDSE or a Landsat scene from EarthExplorer. Options: True/False

scene_download_directory=None
Directory where to download CDSE/EarthExplorer scenes. Set to output setting if None.

4.1.4. Various processing

min_tgas_aot=0.85

Minimum gas transmittance required to use a band for aerosol optical thickness retrieval.

min_tgas_rho=0.70

Minimum gas transmittance required to retrieve surface reflectance for a band.

cirrus_correction=False

Attempt correction of cirrus clouds using one or more cirrus bands in the 1.37 μm range. If multiple bands are present, ρ_{cirrus} will be averaged. Currently all reflectance will be assumed from cirrus, and is hence this correction is not suitable for high elevation targets. Cirrus reflectance at 1.37 μm ($\rho_{cirrus} = \rho_t - \rho_{Rayleigh}$) will be transfered to VNIR and SWIR bands using the factors below. Options=True/False

cirrus_range=1350,1390

Minimum and maximum wavelength for detecting cirrus clouds.

cirrus_g_vnir=1.0

Factor to transfer cirrus reflectance to VNIR bands.

cirrus_g_swir=1.0

Factor to transfer cirrus reflectance to SWIR bands.

blackfill_skip=True

Skip a region of interest when it is outside the sensor swath or in the "blackfill" around a projected satellite image, e.g. the empty borders around Landsat data. Options: True/False

blackfill_max=1.0

Fraction of "blackfill" in the region of interest to skip processing. For example, a value of 1.0 will skip scenes if the region of interest is fully outside the satellite swath, a value of 0.5 will allow up to 50% of the region of interest to be outside the swath.

blackfill_wave=1600

Band in which to check "blackfill" cover, as identified by wavelength.

*copy_datasets=lon,lat,rhot_**

Datasets that are copied from L1R to L2R NetCDF files. Editing this may break L2W generation.

luts=ACOLITE-LUT-202110-MOD1,ACOLITE-LUT-202110-MOD2

LUTs to use in processing. By default MOD1 (continental) and MOD2 (maritime) from 6SV are included. Can be reduced to a single model if preferred by the user.

luts_pressures=500,750,1013,1100

LUT pressure values to load. Availability of specific pressure datasets depends on the LUT. Can be reduced to a specific pressure range for faster loading.

luts_reduce_dimensions=False

Reduce the range of viewing zenith angles (0-16 degrees) and aerosol optical depth (0-1) in the LUT loading. By default set to False, but it is set to True for Landsat and Sentinel-2 type sensors with limited ranges of viewing zenith angles.

sza_limit_replace=False

Replace sun zenith angles exceeding the values in the LUTs. Not recommended for general use as even the LUT maximum is large.

sza_limit=79.999

Sun zenith angle maximum in the LUT.

4.2. Output options

rgb_rhot=True

Controls the output of RGB PNG files based on ρ_t data. Options: True/False

rgb_rhos=True

Controls the output of RGB PNG files based on ρ_s data. Options: True/False

rgb_rhow=False

Controls the output of RGB PNG files based on ρ_w data. Options: True/False

map_l2w=False

Controls the output of PNG files from the requested L2W parameters. Options: True/False

output_geolocation=True
Controls the output of longitude and latitude to the NetCDF files. Options: True/False

output_xy=False
Controls the output of x and y easting and northing to the NetCDF files. Options: True/False

output_geometry=True
Controls the output of resolved view and illumination geometry (if available) to the NetCDF files. Options: True/False

output_rhorc=False
Controls the output of Rayleigh corrected reflectance to the NetCDF files. Will also be set if any rhorc_* output is listed in l2w_parameters. Options: True/False

output_bt=False
Controls the output of at-sensor brightness temperatures for the Landsat thermal bands to the NetCDF files. Will also be set if any bt* output is listed in l2w_parameters. Options: True/False

output_lt=False
Output top-of-atmosphere radiances for some sensors (e.g. DESIS, HICO, HYPERION, PRISMA). Options: True/False

solar_irradiance_reference=Coddington2021_1_0nm
Reference solar irradiance spectrum to use in top-of-atmosphere radiance to reflectance conversion. Only for sensors where ACOLITE does this conversion. Options: Coddington2021_1_0nm, Coddington2021_native, Gueymard2004, Gueymard2018, Kurucz1992_0_1nm, Kurucz1992_1_0nm, Meftah2017, SAO2010, Thuillier2003, Wehrli1985

netcdf_projection=True
Output projection info to NetCDF so it can be used by gdal. Options: True/False

netcdf_compression=False
Controls whether NetCDF outputs use internal compression. Options: True/False

netcdf_compression_level=4
Compression level for NetCDF internal compression. Options: Integer value between 1 (leastcompression) and 9 (most compression).

netcdf_compression_least_significant_digit=None
Least significant digit for use in the NetCDF internal compression.

netcdf_discretisation=False
Discretise datasets when writing, based on config/parameter_discretisation. Options: True/False

l1r_export_geotiff=False
Controls the output of L1R GeoTIFF files. Options: True/False

l2r_export_geotiff=False
Controls the output of L2R GeoTIFF files. Options: True/False

l2t_export_geotiff=False
Controls the output of L2T GeoTIFF files. Options: True/False

l2w_export_geotiff=False
Controls the output of L2W GeoTIFF files. Options: True/False

delete_acolite_run_text_files=False
Controls whether ACOLITE text files are deleted. Options: True/False

delete_acolite_output_directory=False
Controls whether ACOLITE generated output directory is deleted if empty. Options: True/False

export_geotiff_coordinates=False
Controls the output of coordinate datasets (lon, lat, x, y) to GeoTIFF files. Options: True/False

export_geotiff_match_file=False
Full path to a file to copy the GeoTiFF tag (projection/RPC) data from when outputting GeoTIFF files. Useful if the projection of a scene is not supported by ACOLITE.

export_cloud_optimized_geotiff=False
Output cloud optimised GeoTIFFs - only with netcdf_projection=True. Options: True/False

l1r_export_geotiff_rgb=False
Output L1R RGB GeoTIFF. Uses the stretch values in rgb_min and rgb_max. Options: True/False

l2r_export_geotiff_rgb=False
Output L2R RGB GeoTIFF. Uses the stretch values in rgb_min and rgb_max. Options: True/False

l1r_delete_netcdf=False
Controls the deletion of L1R NetCDF files. Options: True/False

l2r_delete_netcdf=False
Controls the deletion of L2R NetCDF files. Options: True/False

l2r_pans_delete_netcdf=False
Controls the deletion of L2R pans NetCDF files. Options: True/False

l2t_delete_netcdf=False
Controls the deletion of L2T NetCDF files. Options: True/False

l2w_delete_netcdf=False
Controls the deletion of L2W NetCDF files. Options: True/False

reproject_outputs=L1R,L2R,L2W
Controls which NetCDF outputs should be projected to a map projection. Options: comma separated list of output types (L1R, L2R, L2W, L2T).

reproject_before_ac=False
Controls whether to reproject the converted L1R NetCDF before atmospheric correction. Faster since only one NetCDF is reprojected. May give slightly different results due to DSF running on projected/unprojected data. Options: True/False.

output_projection=False
Controls whether outputfiles are projected to a map projection. Options: True/False.

output_projection_name=None
Name to append to the projected outputfiles. If None, _projected will be appended to the standard filename.

output_projection_epsg=None
EPSG code of the target projection. If output_projection_epsg and output_projection_proj4 not provided the scene center coordinates will be used to select a UTM zone.

output_projection_proj4=None
Proj4 string of the target projection, only used if output_projection_epsg is None. If output_projection_epsg and output_projection_proj4 not provided the scene center coordinates will be used to select a UTM zone.

output_projection_resolution=None
Resolution of the output grid in metres. X and Y resolution in comma separated values. E.g. output_projection_resolution=300,300 for OLCI and MERIS Full Resolution data.

output_projection_limit=None
Bounding coordinates of the output grid, decimal degrees in S,W,N,E order. If not set, the limit bounding coordinate values will be used, or the min/max from the latitude and longitude in the NetCDF file. If output_projection_metres is set, then the coordinates need to provided in projected metres in the output_projection_xrange and output_projection_yrange settings.

output_projection_metres=False
Controls whether to use the xrange and yrange limits in projection space. Options=True/False

output_projection_xrange=None
Controls the xrange in projection space, if output_projection_metres is set.

output_projection_yrange=None
Controls the yrange in projection space, if output_projection_metres is set.

output_projection_resolution_align=True
Controls whether the reprojected grid is aligned with the output resolution size in metres.

output_projection_resampling_method=bilinear
Controls reprojection method.

default_projection_resolution=None

Set default projection resolution for a given sensor. Set in sensor specific settings files.

output_projection_fillnans=False

For some projected images lines of zeros are retrieved. Set this option to fill them with interpolated values. May cause issues at swath edge. Options=True/False

output_projection_filldistance=1

Set the fill distance for reprojection.

output_projection_radius=3

Set the radius for reprojection.

output_projection_epsilon=0

Set the epsilon for reprojection.

output_projection_neighbours=32

Set the neighbours for reprojection.

4.3. Pansharpening

pans=False

Controls pansharpened outputs. Currently capable of pansharpening L2R data from Landsat 7/8/9. You should really consider this only for visualisation purposes. Options: True/False

pans_method=panr

Method for pansharpening, using a ratio of pan to its multispectral equivalent. The ratio can be either computed using RGB bands (visr) or using the PAN band alone (panr). Options: panr, visr

pans_output=rgb

Bands to be pansharpened. It is recommended to only use rgb. Options: rgb, all

pans_bgr=480,560,655

Bands to use in visr method and for rgb output. Do not change.

pans_rgb_rhot=True

Output pansharpened rho_t PNG. Options: True/False

pans_rgb_rhos=True

Output pansharpened rho_s PNG. Options: True/False

pans_rgb_rhot=True

Output pansharpened rho_t PNG. Options: True/False

pans_rgb_rhos=True

Output pansharpened rho_s PNG. Options: True/False

pans_export_geotiff_rgb=False

Output pansharpened rho_s GeoTIFF. Options: True/False

pans_sensors=L7_ETM,L8_OLI,L9_OLI

Supported sensors for pansharpening. Do not change.

4.4. Various

slicing=False

Controls slicing of data to finite values during processing, should speed up and improve memory use for sparse (or custom) images. Options: True/False

verbosity=5

Controls the amount of detail in the terminal outputs during processing. (Verbosity levels are not consistently implemented at the moment.)

delete_extracted_input=True

Controls whether input bundles that were extracted by ACOLITE should be deleted. Options: True/-False

4.5. DSF options

dsf_aot_estimate=fixed
: Whether to use the tiled or fixed DSF. Experimental resolved and segmented settings are also available. Options: fixed, tiled, resolved, segmented

dsf_nbands=2
: Number of bands to use in the DSF AOT retrieval, e.g. for computing min or mean AOT

dsf_nbands_fit=2
: Number of bands to use in the DSF path reflectance fitting

dsf_aot_compute=min
: How to compute AOT from the number of bands. Default: min, Options: min, mean, median

dsf_wave_range=400,900
: Comma separated wavelength range in nm of bands to use in the DSF, e.g. for including (400,2500) or excluding (400,900) the SWIR bands from the fitting. Default: 400,900

dsf_exclude_bands=,
: Comma separated list of bands to exclude in the DSF. Names have to match names in the relative spectral response file of the sensor.

dsf_interface_reflectance=False
: Controls the air-water interface reflectance correction. Disabled by default at the moment since the surface reflectance is quite sensitive to (local) wind speed in nadir viewing conditions. Options: True/False

dsf_interface_option=default
: Method to use for the air-water interface reflectance correction. Currently only for all pixels with no distinction between water and non-water. Options: "default" for the OSOAA (Chami et al., 2015) based "sky" reflectance presented in (Vanhellemont, 2020c; Vanhellemont and Ruddick, 2021)

dsf_interface_lut=RSKY-202102
: Version of the LUT to use for the air-water interface correction.

wind=None
: Force the wind speed (m/s) for the air-water interface correction. If None, the default value will be used or the value retrieved from the ancillary data.

wind_default=None
: Default wind speed (m/s) for the air-water interface correction.

dsf_write_aot_550=False
: Controls output of a per pixel (interpolated or resampled) aerosol optical thickness dataset. Note that in fixed processing this is also output as a single float in the NetCDF global attributes. Options: True/False

4.5.1. Fixed DSF options

dsf_fixed_aot=None
: User specified aerosol optical thickness value at 550 nm. Options: None or float value

dsf_fixed_lut=ACOLITE-LUT-202102-MOD2
: User specified aerosol model for the dsf_fixed_aot option. Options: ACOLITE-LUT-202102-MOD1 or ACOLITE-LUT-202102-MOD2

4.5.2. Tiled DSF options

dsf_tile_dimensions=None
: Comma separated tile size in pixels for the tiled DSF processing. Defaults per sensor are set in config/defaults/.

dsf_min_tile_cover=0.10
: Minimum coverage of valid pixels in the individual tiles. Tiles with less coverage (typically at the swath edges) will be skipped and interpolated. Options: fraction between 0 and 1

dsf_min_tile_aot=0.01
: Minimum aerosol optical thickness (at 550 nm) in a tile. Tiles with lower τ_a will be skipped and interpolated. Typically very low τ_a indicate cloud shadow contamination.

dsf_max_tile_aot=1.20
Maximum aerosol optical thickness (at 550 nm) in a tile. Tiles with higher τ_a will be skipped and interpolated. Typically very high τ_a indicate cloud contamination.

dsf_write_tiled_parameters=False
Controls the output of atmospheric parameters to the NetCDF file. Options: True/False

dsf_min_tile_cover=0.10
Minimum coverage of valid pixels in the individual tiles. Tiles with less coverage (typically at the swath edges) will be skipped and interpolated. Options: fraction between 0 and 1

dsf_tile_smoothing=True
Whether to smooth between tiles in tiled processing. Options: True/False

dsf_tile_smoothing_kernel_size=3
Kernel size for smoothing tiles in tiled processing. Options: positive integer

dsf_tile_interp_method=linear
Method for scaling tiles in tiled processing to full scene. Options: linear,nearest,cubic

4.5.3. Glint correction options

dsf_residual_glint_correction=False
Controls the residual glint correction. Options: True/False

dsf_residual_glint_correction_method=default
Residual glint correction method to use. Options: default/alternative

dsf_residual_glint_wave_range=1500,2400
Wavelength range for the alternative residual glint correction method.

glint_force_band=None
Specify the wavelength (nm) where glint reflectance ρ_g is to be estimated. If None, then in each pixel the SWIR band giving the lowest ρ_g at 440 nm will be used.

glint_mask_rhos_band=1600
The wavelength (nm) to determine which pixels need to be glint corrected. The closest band will be selected.

glint_mask_rhos_threshold=0.05
Threshold to determine which pixels need to be glint corrected. Pixels with glint_mask_rhos_band > the threshold will not be corrected. (It is noted that ρ_g at 1.6 μm can be larger than 0.05.)

glint_write_rhog_ref=False
Controls the output of the reference glint reflectance to the NetCDF file. Options: True/False

glint_write_rhog_all=False
Controls the output of the glint reflectance in each band to the NetCDF file. Options: True/False

4.5.4. DSF development options

These are specialised development options which should not generally be changed by the user.

dsf_spectrum_option=intercept
Method to choose the ρ_{dark}. Options: darkest, percentile, intercept

dsf_percentile=1
Percentile to use for the ρ_{dark} percentile selection.

dsf_intercept_pixels=1000
Number of pixels to use for the ρ_{dark} intercept selection.

dsf_minimum_segment_size=1
Minimum segment size for segmented processing.

dsf_allow_lut_boundaries=False
Allow use of the bounding LUT values. Options:True/False

dsf_model_selection=min_drmsd
Method to choose which aerosol model to use. Options: min_drmsd

dsf_filter_rhot=False

Apply percentile filter to top-of-atmosphere data for retrieving the ρ_{dark}. Options: True/False

dsf_filter_percentile=5

Percentile for the filtering of top-of-atmosphere data for retrieving the ρ_{dark}.

dsf_filter_box=5,5

Box size in pixels for the filtering of top-of-atmosphere data for retrieving the ρ_{dark}.

dsf_filter_aot=False

Filter aerosol optical thickness after retrieval using the previous two settings. Options: True/False

dsf_aot_fillnan=True

Fill NAN values for the resolved aerosol optical thickness. Options: True/False

dsf_smooth_aot=True

Apply Gaussian smoothing to the retrieved aerosol optical thickness in resolved processing mode. Options: True/False

dsf_smooth_box=20,20

Box size in pixels for the smoothing of the resolved aerosol optical thickness.

4.6. EXP options

exp_wave1=1600
: First wavelength (nm) for the EXP correction. The closest band will be selected.

exp_wave2=2200
: Second wavelength (nm) for the EXP correction. The closest band will be selected.

exp_alpha=None
: Ratio of the water reflectances in the two used bands. Only used if no SWIR band is used. If None, the similarity spectrum (Ruddick et al., 2006) is used.

exp_alpha_weighted=True
: Controls the band weighting of the similarity spectrum alpha. Options: True/False

exp_epsilon=None
: Fixed aerosol epsilon value in the two used bands. If None, the value is determined from the scene according to Vanhellemont and Ruddick (2015).

exp_gamma=None
: Fixed transmittance ratio value in the two used bands. If None, the ratio of Rayleigh transmittances is used.

exp_fixed_epsilon=True
: Controls whether a fixed or per pixel (only for SWIR/SWIR) epsilon is used. Options: True/False

exp_fixed_epsilon_percentile=50
: Percentile to determine the fixed epsilon over the (sub)scene. Options: value between 0 and 100

exp_fixed_aerosol_reflectance=True
: Controls whether the aerosol reflectance is fixed over the (sub)scene. Options: True/False

exp_fixed_aerosol_reflectance_percentile=5
: Percentile to determine the fixed aerosol reflectance over the (sub)scene. Options: value between 0 and 100

exp_swir_threshold=0.0215
: Non-water threshold for the SWIR1 band (at 1.6 μm), used to determine the scene wide fixed epsilon.

exp_output_intermediate=False
: Determines output of intermediate datasets (i.e. per pixel epsilon and rhoam) from the exponential processing. Options: True/False

4.7. Sensor specific options

4.7.1. Landsat

landsat_output_qa=False
Controls the output of the Quality Assesment bands from Landsat data. Options: True/False

landsat_qa_bands=PIXEL,RADSAT,SATURATION
Controls which QA bands get output to the L1R file. To output QA bands to L2R add them to copy_parameters. To output QA bands to L2W add them to both copy_parameters and l2w_parameters.

4.7.2. Sentinel-2/MSI

s2_target_res=10
Output resolution of Sentinel-2 products in meter. Options: 10, 20, 60

geometry_type=grids_footprint
Method to derive per pixel geometry data for S2. Options: "grids" for simple interpolation of the 5x5 km grids, "grids_footprint" for extrapolation of the 5x5 km grids and subsetting per detector (recommended), and "gpt" use SNAP to compute the geometry. (Configure "snap_directory" in acolite/config/config.txt)

geometry_res=60
Resolution at which to compute the per pixel geometry data. Geometry will be resampled to s2_target_res, so they can be different values. Computing the geometry at 60 m is generally enough (and faster!). Options: 10, 20, 60

geometry_per_band=False
Compute per band rather than band average geometry. This is only important in glint viewing conditions, and then the ancillary wind speed estimate is probably not accurate enough. Can generally be left False. Only for "grids_footprint" option. Options: True/False.

geometry_fixed_footprint=False
Determines whether band specific footprint polygons or fixed footprint polygons are used for the computation of per pixel view geometry. Only for "grids_footprint" option. Options: True/False.

geometry_override=False
Overwrite S2 GPT generated geometry files if found. Only for "gpt" option. Options: True/False.

s2_include_auxillary=False
Output auxillary grib data provided by ESA. Options: True/False.

s2_project_auxillary=True
Project auxillary grib data to ROI or scene. Options: True/False.

s2_write_dfoo=False
Write Sentinel-2 per pixel detector footprint. Options: True/False

s2_write_vaa=False
Write Sentinel-2 per pixel view azimuth angle. Options: True/False

s2_write_saa=False
Write Sentinel-2 per pixel sun azimuth angle. Options: True/False

s2_dilate_blackfill=False
Dilate the Sentinel-2 nodata ("blackfill") mask. Useful when merging tiles with separate receiving stations are merged. Options: True/False

s2_dilate_iterations=2
Number of iterations for the Sentinel-2 nodata ("blackfill") mask dilation. Options: positive integer

4.7.3. Sentinel-3/OLCI

smile_correction=True
Apply SMILE correction as in Bourg et al. (2008). Options: True/False

smile_correction_tgas=True
Apply SMILE correction with gas transmittance correction. Options: True/False

use_tpg=True

Use longitude and latitude tie point grids for computing region of interest subsetting. Options: True/-False

4.7.4. Planet

planet_store_sr=False
Sets whether to output Planet SR data as ACOLITE NetCDF if available. If only Planet SR data is given there will be no further processing. Options: True/False

4.7.5. Pléiades

pleiades_skip_pan=False
Skip Pléiades panchromatic band during processing. Much faster since the panchromatic data (16x the number of pixels compared to multispectral bands!) does not need to be read or written. Options: True/False

4.7.6. Worldview

inputfile_swir=None
Supply WorldView SWIR bundle that corresponds to the VNIR bundle for simultaneous processing.

worldview_reproject=False
Reproject unprojected WV bundles to UTM. Options: True/False

worldview_reproject_resolution=2
Pixel resolution for the WV UTM reprojection

worldview_reproject_method=nearest
Method for the WV UTM reprojection. Options: nearest, bilinear

4.7.7. PRISMA

prisma_rhot_per_pixel_sza=False
Whether to use resolved sun zenith angles grid to compute top-of-atmosphere reflectance (rho_t) from top-of-atmosphere radiance (L_t). Options: True/False

prisma_store_l2c=False
Whether to extract and store PRISMA L2C surface level reflectances (rhos_l2c_*) from the official L2C file. Options: True/False

prisma_store_l2c_separate_file=True
Whether to store PRISMA L2C surface level reflectances (rhos_l2c_*) in a separate converted_L2C file. If False they will be stored in the L1R. Options: True/False

4.7.8. CHRIS

chris_interband_calibration=True
Apply CHRIS top-of-atmosphere interband calibration from Lavigne et al. (2021). Options: True/False

chris_noise_reduction=True
Apply CHRIS noise reduction based on Gómez-Chova et al. (2008). Options: True/False

4.7.9. DESIS

desis_mask_ql =True
Apply masks from the DESIS QL QUALITY dataset. Options: True/False

4.7.10. Gaofen

clear_scratch=False
Clear scratch directory. Options: True/False

gf_reproject_to_utm=False
Attempt to reproject GF imagery to UTM if only provided with RPC tags. Warning: Experimental! Options: True/False

4.7.11. VIIRS

viirs_option=img+mod
Controls the bands output in VIIRS processing. First element denotes the output resolution, either at I or at M size. Options: img+mod,mod+img,img,mod

viirs_scanline_projection=True
Controls whether VIIRS reprojection is done per estimated scan line. Options: True/False

viirs_scanline_width=32
Number of pixels in the VIIRS scan line for the reprojection. Options: 32 or 16

viirs_quality_flags=4,8,512,1024,2048,4096
Which L1B quality flags are used to mask VIIRS data. Options: Comma separated list of integers.

viirs_output_tir=True
Controls whether VIIRS thermal data should be output. Options: True/False

viirs_output_tir_lt=False
Controls whether VIIRS thermal data should be output also as radiance. Options: True/False

viirs_mask_mband=True
Controls masking of VIIRS data using the M bands. Options: True/False

viirs_mask_immixed=True
Controls masking of VIIRS data using common I and M bands. Options: True/False

viirs_mask_immixed_rat=False
Controls masking of VIIRS data using common I and M bands ratio. Options: True/False

viirs_mask_immixed_dif=True
Controls masking of VIIRS data using common I and M bands difference. Options: True/False

viirs_mask_immixed_bands=I03/M10
Controls band pairs used in masking of VIIRS data using common I and M bands. Options: I01/M05,I02/M07,I03/M10

viirs_mask_immixed_maxrat=0.2
Set maximum ratio for masking of VIIRS data using common I and M bands ratio.

viirs_mask_immixed_maxdif=0.002
Set maximum difference for masking of VIIRS data using common I and M bands difference.

4.8. Output algorithm settings

nechad_range=600,900

Wavelength range for Nechad type turbidity and suspended particulate matter algorithms, used to match sensor bands when expanding wildcards (e.g. tur_nechad2009_*).

nechad_max_rhow_C_factor=0.5

Maximum ρ_w to compute Nechad turbidity and suspended particulate matter for, as a factor of the band specific C. A factor of 0.5 limits ρ_w to regions where $\rho_w < 0.5$ C.

4.8.1. Contrabands

compute_contrabands=False

Controls the output of the generalised contrabands (Castagna et al., 2020). Defaults are True for Landsat 8 and 9 OLI and the EO1/ALI. Options: True/False

4.9. TACT settings

tact_run=False

Controls whether the Thermal Atmospheric Correction Tool (TACT) should be run for Landsat thermal data (L5/TM B6, L7/ETM B6, L8/TIRS B10, B11). Options: True/False

tact_profile_source=era5

Source of atmospheric profiles for the Thermal Atmospheric Correction Tool (TACT). Use era5 for scenes older than 90 days, gdas1 for more recent scenes. Options: era5/gdas1/ncep.reanalysis2

tact_emissivity=water

Which emissivity option to use. Options: water/user/unity/ged If ged, ASTER GED data will be used, otherwise a default emissivity json file will be loaded.

tact_emissivity_file=None

Path to a specific emissivity json file to load. Note the formatting of the default emissivity json files in data/TACT

tact_output_atmosphere=False

Controls whether the simulated atmospheric parameters from the Thermal Atmospheric Correction Tool (TACT) should be output to the NetCDF file. Options: True/False

tact_output_intermediate=False

Controls whether the intermediate parameters (at-sensor brightness temperature, radiance, surface radiance, surface emissivity) from the Thermal Atmospheric Correction Tool (TACT) should be output to the NetCDF file. Options: True/False

tact_map=True

Controls whether the TACT outputs should be output as PNG maps. Options: True/False

tact_range=3.5,14

Wavelength range for TACT in microns

4.10. General mapping options

map_title=True
: Controls the title on PNG maps. The title consists of the parameter, satellite sensor and date. Options: True/False

map_fontname=sans-serif
: Controls font used in the title and labels of the output maps.

map_usetex=False
: Controls whether latex fonts are used in the title and labels of the output maps. Options: True/False

map_projected=False
: Use cartopy for mapping. Does not work properly yet, does not work at all in binary version. Datasets will be reprojected and some distortion may occur. Options: True/False

map_raster=False
: Controls whether maps are generated using matplotlib or pillow. Raster maps are 1:1 and have no annotations. Options: True/False

map_pcolormesh=False
: Use pcolormesh rather than imshow for mapping using matplotlib. Options: True/False

map_cartopy=False
: Try using cartopy for map outputs. Options: True/False

map_dpi=300
: DPI of the output maps.

map_ext=png
: File type of the output maps.

map_limit=None
: Crop output maps to this limit (S,W,N,E).

map_scalebar=False
: Controls whether an approximate scalebar is added to the output maps. Options: True/False

map_scalebar_position=UL
: Controls position of the scalebar. Options: UL/TL/BL/BR

map_scalebar_color=Black
: Controls colour of the scalebar. Options: Named colour of HTML colour

map_scalebar_length=None
: Set scalebar length in km. If None then the scalebar length will be determined.

map_scalebar_max_fraction=0.33
: Maximum length of scalebar as fraction of the horizontal dimension of the image. Options: value between 0 and 1.

map_points=None
: Path to a file with points to be annotated on the maps.

4.10.1. RGB mapping options

rgb_red_wl=660
: The wavelength (nm) for the Red channel in the RGB composite. The band with closest wavelength will be selected.

rgb_green_wl=560
: The wavelength (nm) for the Green channel in the RGB composite. The band with closest wavelength will be selected.

rgb_blue_wl=480
: The wavelength (nm) for the Blue channel in the RGB composite. The band with closest wavelength will be selected.

rgb_min=0.0,0.0,0.0
: Comma separated minimum reflectance for scaling of the R,G,B channels.

rgb_max=0.15,0.15,0.15

Comma separated maximum reflectance for scaling of the R,G,B channels.

rgb_gamma=1.0,1.0,1.0

Gamma exponent to be applied to data before RGB stretching.

rgb_autoscale=False

Use percentiles of the data to scale the R,G,B channels individually.

rgb_autoscale_percentiles=5,95

Percentile values for linear scaling of the R,G,B channels when rgb_autoscale is set.

rgb_stretch=linear

Stretching method to go from rgb_min:rgb_max to 0:255 UINT8. Options: linear, log, sinh, sqrt

4.10.2. L2W mapping options

map_colorbar=True

Controls the plotting of colour bars for the L2W PNG maps. Options: True/False

map_colorbar_orientation=vertical

Controls the orientation of the colour bars. Options: horizontal/vertical

map_auto_range=False

Controls the auto ranging for the L2W maps. If False the ranges in config/parameter_labels.txt will be used. Options: True/False

map_auto_range_percentiles=1,99

Lower and upper percentiles for the colour bar auto ranging.

map_outrange=False

Also fill the pixels that are out of the colour bar range. If False the out of range pixels will use the colour from the respective end of the colour bar. Options: True/False

map_fillcolor=LightGrey

Controls the fill colour in the L2W maps. Options: any Python colour name or Hex Color Code

map_default_colormap=viridis

Default colormap to use if dataset is not configured in config/parameter_labels.txt For list of options see matplotlib documentation: https://matplotlib.org/stable/tutorials/colors/colormaps.html

5. L2W algorithms

ACOLITE includes several algorithms for the retrieval of parameters derived from reflectance. The L2R file is used as an input and an L2W file is generated containing the requested datasets. The L2W file will be used to generate maps from the parameters if requested. The map scaling for each parameter is defined in the config/acolite_l2w_labels.txt file. The included output products and algorithms are:

- **rhot_***: ρ_t, the top-of-atmosphere reflectance as derived from the original input file. A wavelength can be specified, or the asterisk will be expanded to include all available wavelengths. Sensors: all sensors
- **rhos_***: ρ_s, the surface level reflectance. A wavelength can be specified, or the asterisk will be expanded to include all available wavelengths. Sensors: all sensors
- **rhow_***: ρ_w, the surface level reflectance for water pixels, with a SWIR based mask applied to non-water pixels. A wavelength can be specified, or the asterisk will be expanded to include all available wavelengths. Sensors: all sensors
- **Rrs_***: the remote sensing reflectance (sr^{-1}) for water pixels, Rrs = ρ_w / π. A wavelength can be specified, or the asterisk will be expanded to include all available wavelengths. Sensors: all sensors
- **rhorc_***: ρ_{rc}, the Rayleigh corrected reflectance. This is the ρ_t with ρ_r removed and corrected for two way Rayleigh transmittance. A wavelength can be specified, or the asterisk will be expanded to include all available wavelengths. Sensors: all sensors
- **spm_nechad2010**: Suspended Matter Concentration (gm^{-3}) using the algorithm of Nechad et al. (2010). By default the red band will be used, but a wavelength can be specified (e.g. spm_nechad2010_833 for S2B) or one of the aliases can be used for the red (spm_nechad2010_red) or NIR (spm_nechad2010_nir) bands. A wildcard is supported to export the parameter for all suitable bands between 600 and 900 nm (spm_nechad2010_*). The parameters from the closest wavelength from the hyperspectral dataset will be used. Sensors: all sensors
- **spm_nechad2016**: Suspended Matter Concentration (gm^{-3}) using the algorithm of Nechad et al. (2010) recalibrated by Bouchra Nechad in September 2016, specifically for Landsat 8 and Sentinel-2. By default the red band will be used, but a wavelength can be specified (e.g. spm_nechad2016_833 for S2B) or one of the aliases can be used for the red (spm_nechad2016_red) or NIR (spm_nechad2016_nir) bands. Sensors: L8/OLI, S2A/MSI, S2B/MSI
- **spm_nechad2010ave**: Suspended Matter Concentration (gm^{-3}) using the algorithm of Nechad et al. (2010). By default the red band will be used, but a wavelength can be specified (e.g. spm_nechad2010ave_833 for S2B) or one of the aliases can be used for the red (spm_nechad2010ave_red) or NIR (spm_nechad2010ave_nir) bands. A wildcard is supported to export the parameter for all suitable bands between 600 and 900 nm (spm_nechad2010ave_*). The hyperspectral dataset will be resampled to the band relative spectral response. Sensors: all sensors **Warning: not recommended for bands extending outside the algorithm's optimal spectral range.**
- **tur_nechad2009**: Turbidity (FNU) using the algorithm of Nechad et al. (2009). By default the red band will be used, but a wavelength can be specified (e.g. tur_nechad2009_833 for S2B) or one of the aliases can be used for the red (tur_nechad2009_red) or NIR (tur_nechad2009_nir) bands. A wildcard is supported to export the parameter for all suitable bands between 600 and 900 nm (tur_nechad2009_*). The parameters from the closest wavelength from the hyperspectral dataset will be used. Sensors: all sensors
- **tur_nechad2016**: Turbidity (FNU) using the algorithm of Nechad et al. (2009) recalibrated by Bouchra Nechad in September 2016, specifically for Landsat 8 and Sentinel-2. By default the red band will be used, but a wavelength can be specified (e.g. tur_nechad2016_833 for S2B) or one of the aliases can be used for the red (tur_nechad2016_red) or NIR (tur_nechad2016_nir) bands. Sensors: L8/OLI, S2A/MSI, S2B/MSI

- **tur_nechad2009ave**: Turbidity (FNU) using the algorithm of Nechad et al. (2009). By default the red band will be used, but a wavelength can be specified (e.g. tur_nechad2009ave_833 for S2B) or one of the aliases can be used for the red (tur_nechad2009ave_red) or NIR (tur_nechad2009ave_nir) bands. A wildcard is supported to export the parameter for all suitable bands between 600 and 900 nm (tur_nechad2009ave_*). The hyperspectral dataset will be resampled to the band relative spectral response. Sensors: all sensors **Warning: not recommended for bands extending outside the algorithm's optimal spectral range.**

- **tur_dogliotti2015**: Turbidity (FNU) using the algorithm of Dogliotti et al. (2015). The switching algorithm uses the red band for $\rho_{red} < 0.05$, and the NIR band for $\rho_{red} > 0.07$, with a linear weighing in between. The red (tur_dogliotti2015_red) and NIR (tur_dogliotti2015_nir) products can also be output separately. Sensors: all sensors **Warning: Published calibration for MODIS is used for all sensors.**

- **tur_novoa2017**: Turbidity (FNU) using the band switching algorithm of Novoa et al. (2017) and turbidity retrieval of Nechad et al. (2009). Sensors: all sensors.

- **spm_novoa2017**: Suspended Matter Concentration (gm^{-3}) using the band switching algorithm of Novoa et al. (2017) and suspended matter concentration retrieval of Nechad et al. (2010). Sensors: all sensors.

- **chl_oc2, chl_oc3**: Chlorophyll a concentration ($\mu g/l$) using the blue/green ratio algorithm. The oc2 and oc3 use two and three bands respectively. Coefficients are provided in data/Shared/algorithms/chl_oc/defaults.txt, and users can add new coefficients in that file. Calibration values are taken from (Franz et al., 2015) or from the NASA/OBPG OceanColor website. Note that these products should be used with care in coastal and inland waters, especially in the presence of sediments and CDOM. Sensors: L8/OLI, L9/OLI, S2A/MSI, S2B/MSI, S3A/OLCI, S3B/OLCI

- **chl_re_gons, chl_re_gons740**: Chlorophyll a concentration ($\mu g/l$) using the red edge algorithm by Gons et al. (2002) with published coefficients and a mass specific chlorophyll a absorption of 0.015. By default 780 nm (band 6) is used as a reference, but the chl_re_gons740 product uses 740 nm (band 5) on MSI. Output products are by default only produced for waters where $\rho_s 664 > 0.005$ and $\rho_s 704/\rho_s 664 > 0.63$ (thresholds defined by Héloïse Lavigne). Sensors: S2A/MSI, S2B/MSI, S3A/OLCI, S3B/OLCI

- **chl_re_moses3b, chl_re_moses3b740**: Chlorophyll a concentration ($\mu g/l$) using the three band red edge algorithm by Moses et al. (2012). By default 780 nm (band 6) is used as a reference, but the chl_re_moses3b740 product uses 740 nm (band 5) on MSI. Sensors: S2A/MSI, S2B/MSI, S3A/OLCI, S3B/OLCI

- **chl_re_mishra**: Chlorophyll a concentration ($\mu g/l$) using the Normalised Difference Chlorophyll Index algorithm by Mishra and Mishra (2012). Sensors: S2A/MSI, S2B/MSI, S3A/OLCI, S3B/OLCI

- **ndci**: The Normalised Difference Chlorophyll Index algorithm by Mishra and Mishra (2012). Sensors: S2A/MSI, S2B/MSI, S3A/OLCI, S3B/OLCI

- **chl_re_bramich**: Chlorophyll a concentration ($\mu g/l$) using an updated version of the Gons algorithm (Gons et al., 2002) for Sentinel-2 by Bramich et al. (2021). Sensors: S2A/MSI, S2B/MSI, S3A/OLCI, S3B/OLCI

- **slh**: The Scattering Line Height algorithm by Kudela et al. (2015). Sensors: S2A/MSI, S2B/MSI, S3A/OLCI, S3B/OLCI

- **fai, fai_rhot**: The Floating Algal Index by Hu (2009). By default surface reflectance (ρ_s) is used. **fai_rhot** uses the top-of-atmosphere reflectance (ρ_t). Sensors: L5/TM, L7/ETM, L8/OLI, L9/OLI, S2A/MSI, S2B/MSI

- **fait**: The Floating Algal Index adapted to Turbid waters by Dogliotti et al. (2018). Surface reflectance (ρ_s) is used instead of Rayleigh corrected reflectance. Sensors: L8/OLI, L9/OLI, S2A/MSI, S2B/MSI

- **ndvi, ndvi_rhot**: Normalised Difference Vegetation Index. By default surface reflectance (ρ_s) is used. **ndvi_rhot** uses the top-of-atmosphere reflectance (ρ_t). Sensors: all sensors

- **qaa, qaa5, qaa6**: Outputs from the Quasi-Analytical Algorithm (QAA) of Lee et al. (2002), from both versions (qaa) or specifically from version 5 (qaa5) or 6 (qaa6). Parameters can be requested separately, see further. <u>Sensors:</u> L8/OLI, L9/OLI, S2A/MSI, S2B/MSI
- **qaa_rrs_443, qaa_rrs_490, qaa_rrs_560, qaa_rrs_665**: Subsurface Remote sensing reflectance from the Quasi-Analytical Algorithm (QAA) of Lee et al. (2002). <u>Sensors:</u> L8/OLI, L9/OLI, S2A/MSI, S2B/MSI
- **u_rrs_443, u_rrs_490, u_rrs_560, u_rrs_665**: u parameter from the Quasi-Analytical Algorithm (QAA) of Lee et al. (2002). <u>Sensors:</u> L8/OLI, L9/OLI, S2A/MSI, S2B/MSI
- **qaa_v5_a_443, qaa_v5_a_490, qaa_v5_a_560, qaa_v5_a_665, qaa_v6_a_443, qaa_v6_a_490, qaa_v6_a_560, qaa_v6_a_665**: absorption outputs from the Quasi-Analytical Algorithm (QAA) of Lee et al. (2002). The v5 or v6 specification denote version 5 or 6 outputs. <u>Sensors:</u> L8/OLI, L9/OLI, S2A/MSI, S2B/MSI
- **qaa_v5_bbp_443, qaa_v5_bbp_490, qaa_v5_bbp_560, qaa_v5_bbp_665, qaa_v6_bbp_443, qaa_v6_bbp_490, qaa_v6_bbp_560, qaa_v6_bbp_665**: particulate backscattering outputs from the Quasi-Analytical Algorithm (QAA) of Lee et al. (2002). The v5 or v6 specification denote version 5 or 6 outputs. <u>Sensors:</u> L8/OLI, L9/OLI, S2A/MSI, S2B/MSI
- **qaa_v5_Kd_443, qaa_v5_Kd_490, qaa_v5_Kd_560, qaa_v5_Kd_665, qaa_v6_Kd_443, qaa_v6_Kd_490, qaa_v6_Kd_560, qaa_v6_Kd_665**: diffuse attenuation outputs from the Quasi-Analytical Algorithm (QAA) of Lee et al. (2002). The v5 or v6 specification denote version 5 or 6 outputs. <u>Sensors:</u> L8/OLI, L9/OLI, S2A/MSI, S2B/MSI
- **qaa_v6_KPAR_Lee, qaa_v6_Zeu_Lee**: diffuse attenuation of photosynthetically available radiation and euphotic zone depth from the Quasi-Analytical Algorithm (QAA) of Lee et al. (2002, 2007). <u>Sensors:</u> L8/OLI, L9/OLI, S2A/MSI, S2B/MSI
- **qaa_v5_KdPAR_Nechad, qaa_v6_KdPAR_Nechad**: Nechad (unpublished?) version 2 fit of Kd PAR to Kd 490 outputs of the Quasi-Analytical Algorithm (QAA) of Lee et al. (2002). The v5 or v6 specification denote version 5 or 6 outputs. <u>Sensors:</u> L8/OLI, L9/OLI, S2A/MSI, S2B/MSI
- **p3qaa**: Output all the parameters from the QAA-RGB (Pitarch and Vanhellemont, 2021). <u>Sensors:</u> L5/TM, L7/ETM, L8/OLI, L9/OLI, S2A/MSI, S2B/MSI, PlanetScope (0c, 0d, 0e, 0f, 2x), Pléiades-1A, Pléiades-1B, RapidEye, SPOT6, SPOT7, Venμs, WorldView2, WorldView3
- **p3qaa_a_***: Total absorption from the QAA-RGB (Pitarch and Vanhellemont, 2021). R/G/B wavelength can be specified. <u>Sensors:</u> L5/TM, L7/ETM, L8/OLI, L9/OLI, S2A/MSI, S2B/MSI, PlanetScope (0c, 0d, 0e, 0f, 2x), Pléiades-1A, Pléiades-1B, RapidEye, SPOT6, SPOT7, Venμs, WorldView2, WorldView3
- **p3qaa_bb_***: Total backscattering from the QAA-RGB (Pitarch and Vanhellemont, 2021). R/G/B wavelength can be specified. <u>Sensors:</u> L5/TM, L7/ETM, L8/OLI, L9/OLI, S2A/MSI, S2B/MSI, PlanetScope (0c, 0d, 0e, 0f, 2x), Pléiades-1A, Pléiades-1B, RapidEye, SPOT6, SPOT7, Venμs, WorldView2, WorldView3
- **p3qaa_Kd_***: Total Kd from the QAA-RGB (Pitarch and Vanhellemont, 2021). R/G/B wavelength can be specified. <u>Sensors:</u> L5/TM, L7/ETM, L8/OLI, L9/OLI, S2A/MSI, S2B/MSI, PlanetScope (0c, 0d, 0e, 0f, 2x), Pléiades-1A, Pléiades-1B, RapidEye, SPOT6, SPOT7, Venμs, WorldView2, WorldView3
- **p3qaa_zSD**: Secchi depth with bias removed from the QAA-RGB (Pitarch and Vanhellemont, 2021). <u>Sensors:</u> L5/TM, L7/ETM, L8/OLI, L9/OLI, S2A/MSI, S2B/MSI, PlanetScope (0c, 0d, 0e, 0f, 2x), Pléiades-1A, Pléiades-1B, RapidEye, SPOT6, SPOT7, Venμs, WorldView2, WorldView3
- **p3qaa_zSD_biased**: Biased Secchi depth from the QAA-RGB (Pitarch and Vanhellemont, 2021). <u>Sensors:</u> L5/TM, L7/ETM, L8/OLI, L9/OLI, S2A/MSI, S2B/MSI, PlanetScope (0c, 0d, 0e, 0f, 2x), Pléiades-1A, Pléiades-1B, RapidEye, SPOT6, SPOT7, Venμs, WorldView2, WorldView3

- **p3qaa_eta**: eta parameter from the QAA-RGB (Pitarch and Vanhellemont, 2021). Sensors: L5/TM, L7/ETM, L8/OLI, L9/OLI, S2A/MSI, S2B/MSI, PlanetScope (0c, 0d, 0e, 0f, 2x), Pléiades-1A, Pléiades-1B, RapidEye, SPOT6, SPOT7, Venμs, WorldView2, WorldView3
- **p3qaa_KPAR**: KPAR from (Lee et al., 2018) as derived from the QAA-RGB zSD Secchi depth (Pitarch and Vanhellemont, 2021). Sensors: L5/TM, L7/ETM, L8/OLI, L9/OLI, S2A/MSI, S2B/MSI, PlanetScope (0c, 0d, 0e, 0f, 2x), Pléiades-1A, Pléiades-1B, RapidEye, SPOT6, SPOT7, Venμs, WorldView2, WorldView3
- **bt***: At-sensor brightness temperature from the Landsat sensors. Individual parameters listed below. Sensors: L5/TM, L7/ETM, L8/OLI, L9/OLI
- **bt6**: At-sensor brightness temperature from the Thematic Mapper (TM) on Landsat 5. Sensors: L5/TM
- **bt6_vcid_1, bt6_vcid_2**: At-sensor brightness temperature from the Enhanced Thematic Mapper (ETM) on Landsat 7 for both low (1) and high (2) gain levels. The low gain data is less likely to saturate over hot targets. Sensors: L7/ETM
- **bt10, bt11**: At-sensor brightness temperature from the Thermal Infrared Sensor (TIRS) on Landsat 8 and 9. Sensors: L8/TIRS, L9/TIRS,
- **hue_angle**: The Hue Angle (°) by van der Woerd and Wernand (2018). Sensors: L8/OLI, S2A/MSI, S2B/MSI
- **rhos_592, rhow_592, rhos_594, rhow_594**: Panchromatic band reflectance (rhos_592, rhos_594 for L8 and L9). These are now output by default, record kept here for reference. Sensors: L8/OLI, L9/OLI
- **rhos_613, rhow_613**: Orange 613 nm *contra*-band reflectances by Castagna et al. (2018, 2020). These are now output by default, record kept here for reference. Sensors: L8/OLI, L9/OLI
- **olh**: Orange *contra*-band line height by Castagna et al. (2018, 2020). Sensors: L8/OLI

6. Sensor wavelengths

ACOLITE computes the wavelengths of each sensor by averaging the RSR. Here a list of wavelengths is provided per sensor. Note that bands with low gas transmittance (<0.75) are skipped for ρ_s computation. Bands in italic here have lower gas transmittance for *italic* here for zenith sun, nadir viewing and default ozone and water vapour. Hyperspectral sensor wavelengths (CHRIS, DESIS, ENMAP, PRISMA) are taken from the inputfile metadata. For VIIRS the first three wavelengths represent the imaging "I" bands. VIIRS reflectance outputs will include "I" and "M" in the parameter names to distinguish same wavelength bands.

- **AMAZONIA1/WFI**: 490, 558, 652, 821
- **CBERS4A/WFI**: 489, 559, 651, 821
- **CBERS4/WFI**: 491, 560, 664, 815
- **EN1/MERIS**: 413, 443, 490, 510, 560, 620, 665, 681, 709, 754, *762*, 779, 865, 885, 900
- **EO1/ALI**: 592, 442, 485, 567, 660, 790, 866, 1244, 1640, 2225
- **FORMOSAT5/RSI**: 590, 489, 577, 672, 812
- **GeoEye1**: 630, 485, 548, 676, 841
- **GF1D/PMS**: 494, 556, 661, 825, 712
- **GF1/WFV1**: 483, 554, 657, 824
- **GF1/WFV2**: 488, 558, 657, 821
- **GF1/WFV3**: 486, 564, 665, 822
- **GF1/WFV4**: 484, 559, 666, 828
- **GF6/PMS**: 491, 565, 659, 820, 675
- **GF6/WFV**: 488, 558, 659, 826, 701, 749, 432, 609
- **IKONOS2**: 725, 496, 559, 666, 792
- **JPSS-1/VIIRS**: 643, 867, 1604, 411, 445, 489, 557, 667, 746, 868, 1239, *1375*, 1604, 2259
- **JPSS-2/VIIRS**: 642, 868, 1614, 411, 445, 488, 555, 672, 747, 868, 1241, *1382*, 1614, 2252
- **KX10/MII**: 401, 438, 495, 553, 657, 776, 854
- **L1/MSS**: 553, 653, 749, 914
- **L2/MSS**: 550, 661, 752, 910
- **L3/MSS**: 545, 655, 744, 908
- **L4/MSS**: 551, 650, 754, 927
- **L5/MSS**: 553, 650, 757, 931
- **L5/TM**: 486, 571, 660, 839, 1678, 2217
- **L7/ETM**: 479, 561, 661, 835, 1650, 2208, 720
- **L8/OLI**: 443, 483, 561, 655, 865, 1609, 2201, 592, *1373*

- **L9/OLI**: 443, 482, 561, 654, 865, 1608, 2201, 594, *1374*
- **PHR1A**: 501, 561, 650, 835, 657
- **PHR1B**: 505, 558, 663, 844, 661
- **PNEO3**: 439, 486, 563, 655, 723, 826, 643
- **PNEO4**: 439, 486, 563, 655, 724, 826, 644
- **PlanetScope/0c**: 492, 542, 622, 813
- **PlanetScope/0d05**: 493, 542, 621, 812
- **PlanetScope/0d06**: 493, 542, 621, 812
- **PlanetScope/0e**: 518, 552, 633, 812
- **PlanetScope/0f**: 505, 546, 625, 809
- **PlanetScope/22**: 492, 566, 666, 866
- **PlanetScope/SD5**: 492, 566, 666, 707, 866
- **PlanetScope/SD8**: 444, 492, 533, 566, 612, 666, 707, 866
- **QuickBird2**: 718, 492, 551, 654, 810
- **RapidEye**: 477, 556, 658, 709, 804
- **S2A/MSI**: 443, 492, 560, 665, 704, 740, 783, 833, 865, *945*, *1373*, 1614, 2202
- **S2B/MSI**: 442, 492, 559, 665, 704, 739, 780, 833, 864, *943*, *1377*, 1610, 2186
- **S3A/OLCI**: 400, 412, 443, 490, 510, 560, 620, 665, 674, 682, 709, 754, *762*, *765*, 768, 779, 865, 884, *899*, *939*, 1016
- **S3B/OLCI**: 401, 412, 443, 490, 510, 560, 620, 665, 674, 681, 709, 754, *762*, *765*, 768, 779, 865, 884, 899, *939*, 1016
- **Skysat1**: 482, 555, 648, 798, 635
- **Skysat2**: 482, 555, 648, 798, 635
- **Skysat3**: 479, 550, 642, 801, 636
- **Skysat4**: 479, 550, 642, 801, 636
- **Skysat5**: 475, 551, 642, 807, 640
- **Skysat6**: 475, 551, 642, 807, 640
- **Skysat7**: 475, 551, 642, 807, 640
- **Skysat8**: 475, 550, 642, 804, 636
- **Skysat9**: 475, 550, 642, 803, 636
- **Skysat10**: 475, 550, 642, 805, 637
- **Skysat11**: 475, 550, 642, 803, 635

- **Skysat12**: 475, 550, 642, 805, 636
- **Skysat13**: 475, 550, 642, 805, 637
- **Skysat14-Skysat19**: 480, 554, 646, 806, 640
- **SPOT6**: 487, 557, 659, 813, 596
- **SPOT7**: 491, 566, 666, 816, 597
- **SUOMI-NPP/VIIRS**: 638, 862, 1601, 411, 444, 486, 551, 671, 745, 862, 1238, *1375*, 1602, 2257
- **VENμS/VSSC**: 424, 447, 492, 555, 620, 620, 666, 702, 741, 782, 861, *909*
- **WorldView2**: 428, 479, 548, 608, 659, 724, 828, 923, 645
- **WorldView3**: 427, 482, 547, 605, 660, 723, 827, 922, 1210, 1572, 1661, 1730, 2164, 2203, 2260, 2330, 643

7. Version history

20231023.0

– Added support for merging of full S2 tiles, e.g. those from different receiving stations
– Added S2 detector footprint outputs
– Added ENMAP processing
– Added IKONOS2 processing
– Added GF1 WFV processing
– Added MSS processing (Landsat 1–5)
– Added L4/TM processing (using L5/TM LUTs)
– Added SDGSAT-1 KX10/MII processing
– Added EMIT processing
– Added processing of DIMAP subsets of Sentinel-3 OLCI data
– Added QA band output for Landsat
– Added option to change F0 reference
– Added RGB stretching options and gamma
– Added scalebar and points annotation on maps
– Fixed TPG interpolation for Sentinel-3
– Added generalised contrabands
– Added support for PRISMA L1G files
– Added VIIRS processing
– Added support for Planet NTF files
– Fixed Skysat bundle merging
– Added NetCDF discretisation
– Improved nan filling in reprojection algorithm
– Added Novoa like blending algorithm for Nechad TUR and SPM (TUR_Novoa2017 and SPM_Novoa2017)
– Changed settings split
– Added additional Copernicus DEM sources
– Added CDSE and EarthExplorer query and download option
– Changed relative azimuth for DESIS and Pléiades
– New ancillary data options

20221114.0

– Fixed OLCI smile correction using per pixel solar_flux twice

- Changed determination of sensor from OLCI/MERIS data bundles
- Fixed map_projected outputs on Windows
- Added interpolation options for tiled DSF
- Added subsetting for projected WorldView data
- Added support for converting Planet SR data
- Added support for Planet zip bundles with multiple images
- Added font options for output maps
- Added beta version of PNEO3 processing

20221025

- Fixed issues with OLCI gas transmittance correction being applied twice
- Updated OLCI processing defaults, added EUMETSAT SVC, gains=True by default
- Updated PlanetScope processing defaults
- Added new DSF AOT filtering options: mean and median for a given number of (darkest) bands
- Added new DSF path reflectance fitting criteria for a given number of (darkest) bands
- Fixed TOA filtering for AOT determination
- Reduced memory use of glint correction
- Reduced memory use of L2W computation with option l2w_data_in_memory=False
- S2/MSI VAA and SAA data can be output
- Improved ACOLITE directory handling
- Added RGB autorange and output of rhow/rhorc based RGB images
- Added GeoEye processing
- Added FORMOSAT5 processing
- Added support for zipped input bundles
- Added output or PRISMA L2C reflectances in ACOLITE format
- Added support for Copernicus GLO-30 and GLO-60 DEMs
- Added support for SRTM 30m DEM
- Added tile lists for SRTM
- Added reprojection before atmospheric correction
- Improved reprojection speed
- Removed astropy dependency
- Added pan sharpening for L7/L8/L9
- Added TACT support for GDAS1 and ncep.reanalysis2 profiles

- Added TACT ECOSTRESS processing
- Added additional emissivity options to TACT (NDVI, GED, EMINET)
- Skip TACT when ROI is in blackfill

20220222

- Renamed setting l8_orange_band to oli_orange_band
- Fixed issue where the GUI would retain "hidden" settings
- Added EN1/MERIS processing (using 4th reprocessing .SEN3 input format)
- Added EO1/ALI processing
- Added QuickBird2 processing
- Added SuperDove 5B and 8B processing
- Added L9/OLI processing, added L9/TIRS processing
- Added polygon_limit and polylakes options
- Added segmented option for DSF
- Added output of RGB GeoTIFF files
- Added NetCDF projection and NetCDF compression options
- Added option to delete acolite_run text files
- Added srtm15plus option for DEM
- Added output projection options for e.g. projection satellite swath data (OLCI/MERIS data)
- Added RGB GeoTIFF output option
- Added beta processing for AMAZONIA1 and CBERS4A
- Added beta processing for Skysat

20211124

- Fixed issue with default glint correction for fixed geometry and fixed τ_a estimate
- Fixed issue with blue band in the CHL_OC2 algorithm
- Removed wavelength guess for BT datasets
- Added DESIS/HICO/HYPERION processing
- Added Gaofen-6 WFV and PMS processing, added Gaofen-1D PMS processing
- Added support for Sentinel-2 PB004 data
- Added NetCDF removal options
- Fixed rhorc_* output for hyperspectral sensors
- Renamed tur_nechad to tur_nechad2009 and spm_nechad to spm_nechad2010
- Renamed tur_dogliotti to tur_dogliotti2015

– Added tur_nechad2009_* and spm_nechad2010_* wildcards to L2W

20210802

– New generic code base that includes processing of metre-scale sensors (among others Pléiades and PlanetScope) and S3A/B OLCI

– Processing of CHRIS and PRISMA hyperspectral data, Venμs data

– Option to use resolved pressure derived from SRTM DEM

– New per-pixel geometry for Sentinel-2 data, support for per-band per-pixel geometry for Sentinel-2

– Separate repository for LUT storage (acolite/acolite_luts)

– Use of "reverse" LUTs to go from rhot -> aot when estimating aot from resolved geometry

– Panchromatic and orange band (Castagna et al., 2018, 2020) reflectance are output by default for L8/OLI

– Scene merging from different UTM zones

– Polygon (e.g. geojson) ROI definition for projected imagery (e.g. Sentinel-2 and Landsat data)

– Sky and sun glint reflectance can be estimated from (ancillary) wind speed

– Added three band RGB QAA algorithm (Pitarch and Vanhellemont, 2021)

– The following setting names have been changed: dsf_path_reflectance -> dsf_aot_estimate, sky_correction -> dsf_interface_reflectance, sky_correction_option -> dsf_interface_option, sky_correction_lut -> dsf_interface_lut, glint_correction -> dsf_residual_glint_correction

20210114

– Fixed gdal issue for GeoTIFF outputs

20210106

– Update to latest GitHub codebase

– Added options presented in (Vanhellemont, 2020c): removing SWIR bands from aerosol fitting and OSOAA based sky reflectance correction

– Fixed GUI ancillary download issue

– Processing from the Windows GUI can now also be interrupted

– Added options to delete NetCDF files that are produced

– Added support for Collection 2 Landsat 8 data

20190326

– Added correct xrange and yranges to NetCDF global attributes

– Changed Landsat Y subsetting

– Improved Pan band support for merged tiles and orange band computation

– Added support for LO8 files (without TIRS)

– Added strip function to l2w_parameter names in mapping script

– Added some CF convention parameter attributes
– Fixed S2 band naming error during tile merging
– Fixed l2_flags error when masking is disabled
– Fixed plotting of L8 dark spectrum
– Fixed time parsing issue for Landsat files
– Disabled printing of numpy Runtime Warnings

20181210

– Added orange band computation for Landsat 8/OLI, including panchromatic bands output at 30 m
– Added oxygen transmittance
– Added elevation input option
– Added/Fixed SRTM DEM option for elevation computation, missing DEM tiles are printed out
– Added the option to use the resolved S2 view/illumination grids
– Changed import of some Matplotlib features, hopefully allowing map output for EXP processed scenes
– Changed but did not fix download of ancillary data when using the GUI
– Fixed FAIT output on Mac (binary distribution was missing the skimage dependency)
– Fixed output of cirrus band (at TOA)
– Fixed MSI grid naming issue when merging tiles
– Fixed transmittance ratio in glint correction
– Fixed the EXP aerosol correction when L1R NetCDF files are present

20180925

– Explicitly added file encoding to avoid 'file not recognised' errors

20180917

– Added FAIT parameter
– Added glint correction for the DSF
– Added –nogfx option for –cli runs without graphical environments
– Changed relative path identification so the acolite binary can be launched from anywhere
– Fixed blue band bug after chl_oc3 computation
– Fixed ancillary data download in Windows binary distribution
– Fixed some settings file parsing errors
– Fixed default scalebar position for projected maps

20180611

– Added simple Rayleigh corrected reflectance product (with no gas and sky correction): rhorc = (rhot - rhor)/ttr

– Added support for GeoTIFF outputs of L2R/L2W datasets. NetCDF files are still generated, datasets will be also exported as GeoTIFF (l2r_export_geotiff=True, l2w_export_geotiff=True)

– Added optional output of UTM x/y easting and northing coordinates (xy_output=True)

– Added colour table options (Calgae255 and Planck_Parchment_RGB) and possibility for user defined colour tables

– Added output of l2_flags to the L2W file, flags are currently based on SWIR threshold mask (bit 1) and negative rhos (bit 2)

– Added option to disable masking of non-water/bad pixels (l2w_mask_water_parameters=False)

– Added option to output 1:1 pixel PNG files; map_raster=True (without title, geolocation or colourbar, currently not on Windows)

– Added check for rhot < rhor in DSF processing (especially for L5)

– Added pansharpened RGB outputs for L7 and L8

– Added aliases for t_nechad_red and t_nechad_nir products

– Changed naming of RGB output products to rgb_rhot and rgb_rhos

– Sentinel-2 wavelengths are now determined from the RSR instead of the provided metadata. This gives consistent naming before and after the Dec 2017 / Jan 2018 RSR update.

– Fixed the crash when no ancillary data is found or when the oceandata server is unreachable. Fallback values will be used if ancillary data access fails.

– Fixed sub pixel geolocation error in the computation of latitude and longitude: Sentinel-2 scene extents are considered to be at pixel edges, Landsat extents at pixel centres, all ACOLITE output coordinates are reported at pixel centre

– Fixed bug when multiple files are present in the GRANULE/IMG_DATA with the same band name (check if files end in jp2)

– Fixed the plotting of Sentinel-2 dark spectrum fitting results

20180419

– Fixed full tile Sentinel-2 geolocation

– Fixed Sentinel-2 relative azimuth within 0-180°

– Datasets are now processed and stored as 32 bit floats, using less RAM and disk space

– Added rhow (rhos with non-water masking) and Rrs (ρ_w / π) datasets

– Wavelengths are now tracked as attributes in the L2W products (rhot, rhos, rhow, Rrs), allowing for the use of the Spectrum Viewer in SNAP

– Skipping unrecognised and dot files in the .SAFE "GRANULE" and "IMG_DATA" directories

– Added support for autoscaling the L2W output maps: add *,* for the range in config/acolite_l2w_labels.txt

– Added new 'map_projected' option to annotate scalebar and add lat/lon to maps, switched to 'Agg' backend in matplotlib

– Relaxed constrains on the band selection for the chl_oc algorithms (for chl_oc3 outputs of S2A/B)

– Added Hue Angle algorithm

20180327 First public beta version of ACOLITE Python.

– Fixed PNG outputs from GUI

– Fixed Landsat 8 bt10 and bt11 outputs from merged tiles

– Added chl_re_gons configuration file and masking

20180312 First internal beta version of ACOLITE Python.

– Implementation is now in Python (3.6)

– New aerosol correction (DSF) with new lookup tables for path reflectance resampled to each of the supported sensors (L5/TM, L7/ETM, L8/OLI, S2A/MSI, S2B/MSI)

– The new spectral response function of Sentinel-2A is now used

– A more robust tile merging option (including Landsat tiles) has been added

20170718 Last release of ACOLITE IDL.

8. Known issues

- **High RAM use when processing full scenes** Processing full scenes is currently very RAM intensive, especially for operations using several bands, such as computing QAA parameters or generating RGB composites. The system may become unresponsive when low on RAM. ACOLITE may segfault if it runs out of RAM when generating the very large RGB composite.
- **Crash when generating full tile RGB composites** When generating full tile RGB composites (especially for Sentinel-2) the program may segfault, due to running out of RAM, or overriding PNG size limits.
- **Large output files** When processing full scenes, very large output files will be generated. For a full tile of Sentinel-2 data at 10 metres file sizes are about 10 GB for the L1R, and about 12 GB for the L2R files. L2W file size depends on the number of requested parameters (about 0.5 GB/parameter + latitude and longitude).
- **High RAM use when processing hyperspectral data** Processing of hyperspectral data requires the loading of the generic LUTs, this uses plenty of RAM, and processing may fail after the "Loading LUTs" step if not enough RAM is available. At least 16GB of RAM is recommended for processing of hyperspectral data.
- **Slow loading of generic LUTs when processing hyperspectral data** The generic LUTs are retrieved as .bz2 files from the acolite_luts repository. These files are extracted by ACOLITE each run if the extracted .nc file is not available. The loading of the LUTs can be sped up by extracting the .bz2 files manually before running.
- **PRISMA processing requires both L1 and L2C files** Processing of PRISMA data requires both official L1 (PRS_L1_STD_OFFL) and L2C (PRS_L2C_STD) files since the view geometry is only provided in the L2C. The L1 needs to be provided as inputfile, and the L2C needs to be present in the same directory as the L1.

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